Finding Vision-Behavior Circuit through Connectomics

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Summary. Active animals must detect local object motion while suppressing the coherent optic flow produced by their own movement. We propose that the adult *Drosophila melanogaster* brain solves this problem through a retinotopic LLPC1 gateway that converts normalized local motion into flight course-control output. A VCH-gated T4/T5 population supplies local motion to a 100-cell cholinergic LLPC1 sheet. The figure-ground separation is performed *upstream* of LLPC1: VCH pools wide-field T4/T5 motion and feeds GABAergic inhibition back onto the T4/T5 terminals that drive LLPC1, positioned to suppress common-mode motion at those terminals, so that the signal reaching the sheet is already largely separated. LLPC1 is the first retinotopic projection-neuron sheet that reads out this VCH-gated local signal; sheet-wide LPi15 inhibition and recurrent PVLP011 feedback regulate the sheet's operating point and output gain but are not required for the first-pass separation. Descending and whole-CNS motor analyses specify the behavioural meaning of this architecture. Among all LLPC1 targets, the cholinergic cell type Nod1 is the dominant excitatory readout, and it relays the figure signal almost entirely (97%) to steering descending neurons — principally the wing-steering command neuron DNp26 — whereas a weaker PLP/PVLP broadcast route provides the only bridge to jump/escape musculature. Whole-CNS (MCNS) motor mapping confirms that the Nod1 and direct channels drive wing-steering, haltere and neck/gaze systems rather than escape. Cable-distance synapse mapping further places VCH inhibition $\sim 2\,\mu$m from the T4 output terminal that drives LLPC1 (presynaptic gating) and separates dendritic inhibition (LPi15, VCH) from axonal recurrent inhibition (PVLP011) on the sheet. Thus VCH-gated T4/T5 terminals create the residual-like signal and LLPC1 is the retinotopic sheet that reads it out, routing the result mainly to wing-steering course-control descending neurons — several of which drive the steering muscles of a specific wing — rather than a looming/escape pathway.

Introduction

Relative motion is a powerful cue for breaking camouflage. A textured object that is invisible against a matched background becomes salient when it moves with a different velocity or phase from the surrounding panorama. In flies, classical figure-ground experiments showed that textured figures can be detected and tracked against textured grounds, that the effect depends strongly on relative phase, and that small figures can compete behaviorally with much larger backgrounds because the visual system normalizes motion signals across stimulated area [1–4]. Modern *Drosophila* experiments have sharpened the same distinction: object tracking, saccades and figure fixation can be separated from smooth optomotor stabilization, and are supported by parallel visual streams [5–12]; for review see [13].

The elementary motion stage is well defined. T4 and T5 neurons encode ON- and OFF-edge motion and segregate by preferred direction into the four layers of the lobula plate [14–21]. These direction-selective signals are relayed by columnar medulla and lobula neurons [22–25], drive wide-field tangential cells [26–30], and recruit inhibitory lobula-plate networks that implement opponency and gain control [31–33]. What remained unclear was the next transformation: where does a population of local motion detectors become a spatially localized figure signal, and how does that signal reach descending motor control?

Synapse-resolution connectomics turns this from a named-cell puzzle into an architectural one. The adult FlyWire connectome and its cell-type annotations provide the circuit graph needed to ask which pathway simultaneously receives T4/T5 motion, preserves retinotopy, rejects common motion, and reaches premotor output [34–36]. Here we trace this pathway through four connectomic analyses, all performed in the FlyWire and male whole-CNS connectomes. We first map the complete T4/T5 connectivity of the centrifugal horizontal cell VCH and show that it forms a dense reciprocal loop with horizontal T4/T5 channels. We then follow the VCH-gated T4/T5 population to a 100-cell cholinergic LLPC1 sheet, and show that this sheet — and not the other T4/T5 targets — has the properties of a first retinotopic figure-output map. We next enumerate the descending neurons (DNs) that LLPC1 reaches, through a direct channel, a Nod-type central-brain relay and a PLP/PVLP broadcast route. Finally, we map the motor targets of those DNs in the male whole-CNS connectome (MCNS), which contains both brain and ventral nerve cord, and find that the dominant output is wing-steering-led course control.

Here we synthesize these results into a two-stage circuit model (Fig. 1). In the first stage, VCH performs the background suppression, presynaptically gating the T4/T5 terminals that feed LLPC1, so that common-mode motion is removed before the sheet is driven. In the second stage, the cholinergic LLPC1 sheet reads out this VCH-gated local signal retinotopically and routes it to descending motor control; sheet-wide LPi15 inhibition and recurrent PVLP011 feedback regulate the sheet rather than performing the separation. Nod-type neurons and DNs are output relays, not the first figure cells.

Results

The VCH–T4/T5 reciprocal loop is the circuit's entry point

Our analysis begins not with LLPC1 but with VCH, because the centrifugal horizontal cells have long been implicated in motion-defined figure–ground processing: figure-detection (FD) cells in the blowfly are selective for small moving objects [37], and photoinactivation of VCH abolishes that small-field selectivity [38, 39]. We therefore first mapped, at synapse resolution, the complete T4/T5 connectivity of the Left VCH neuron (root 720575940627706398), and used its downstream broadcast to identify which cells are worth following (Fig. 2).

VCH is a GABAergic lobula-plate tangential cell whose dominant input is motion: of its 27,576 input synapses, 12,301 (44.6%) arrive from 1,022 T4/T5 neurons, almost entirely the horizontal T4a/T5a subtypes. VCH in turn synapses back onto 1,476 T4/T5 neurons (7,273 synapses), and 912 of these are *reciprocal* — they both drive VCH and receive its inhibition (89% of VCH's T4/T5 inputs; 62% of its T4/T5 outputs). The excitatory drive (mean 12.0 synapses per partner) is about 2.4 times the inhibitory feedback (mean 4.9), consistent with a graded, proportional feedback loop rather than an all-or-none silencing connection; whether it implements divisive normalization, a canonical computation across sensory systems [40–42], remains to be tested physiologically. Both the input and the output T4/T5 lie in the same (right) optic lobe, so this is a recurrent intra-hemispheric normalization loop, not a cross-hemisphere relay.

The reciprocal T4/T5 population is also a broadcast hub. The 912 neurons make 475,301 output synapses onto 111,815 postsynaptic segments (only ∼12,600 of which are annotated neurons; the remainder are small unproofread fragments, so this is a partner-segment count, not a neuron count). Their dominant identified sinks are lobula-plate intrinsic neurons (LPi14), VCH itself, CT1, the HS family and DCH and — among projection-neuron *populations* — LLPC1 (Fig. 2b). LLPC1 is the target that uniquely combines local T4/T5 motion input with the shared inhibition and central-brain projection expected of a first figure-output sheet, and the remainder of this paper follows it. Full per-partner tables are given in the Supplementary Appendix.

VCH is the only inhibitory T4/T5 target that reaches the figure sheet

The reciprocal T4/T5 population also feeds many large inhibitory neurons (Fig. 2b), and a candidate gate cannot be singled out merely because it is inhibitory and receives motion. The normalization limb of a figure circuit must satisfy three connectomic criteria: it should be *reciprocal* with the T4/T5 population it inhibits; it should directly contact the LLPC1 output sheet; and it should be a centrifugal feedback cell rather than an optic-lobe-intrinsic interneuron. We screened the eight largest inhibitory T4/T5 targets against all three (Fig. 3; Table 1).

Each alternative fails at least one criterion, and they fail in two characteristic ways. The *pan-retinal* inhibitors are reciprocal with the motion field but have no access to the sheet: CT1 is reciprocal with essentially the entire field (5,972 of the ∼6,000 right-hemisphere T4/T5) yet contacts only 3 of the 100 LLPC1, and its columnar terminals are in any case electrically near-isolated compartments [43]; the lobula-intrinsic Li14 and LT33 likewise touch only 14 and 7 of the sheet. The *sheet-blanketing* interneurons are the opposite case: LPi14, LPi15 and the amacrine Am1 contact 90–100 of the 100 LLPC1 but are optic-intrinsic and only weakly reciprocal with T4/T5 (0.56–0.64 of their T4/T5 input is fed back), the signature of feed-forward opponent inhibition rather than a recurrent gain-control loop [32, 33] — consistent with the directional, opponent feed-forward role we assign LPi15 below.

Only the two centrifugal horizontal cells, VCH and DCH, satisfy all three criteria: they are reciprocal with the same T4/T5 they inhibit (912 and 942 partners), directly inhibit the LLPC1 sheet (77 and 52 of 100), and are centrifugal feedback cells rather than optic-intrinsic interneurons. VCH is the stronger of the pair, and adding its presynaptic targeting of the T4/T5→LLPC1 output terminals (shown below) completes a unique anatomical signature. This matches the classical blowfly result that VCH, but not DCH or HSE, is causally necessary for the small-field selectivity of FD cells [38].

Table 1: Inhibitory-normalizer screen of the eight largest inhibitory T4/T5 targets. Reciprocal = number of right-hemisphere T4/T5 that both drive and receive from the cell; LLPC1 contact = number of the 100 LLPC1 it inhibits. Only VCH and DCH are centrifugal, T4/T5-reciprocal and sheet-contacting.

Neuron	Super-class	T4/T5 reciprocal	Recip. fraction	LLPC1 contact (/100)
VCH	`visual_centrifugal`	912	0.89	**77**
DCH	`visual_centrifugal`	942	0.86	52
CT1	`optic`	5,972	1.00	3
LPi15	`optic`	1,771	0.64	100
LPi14	`optic`	1,736	0.59	90
Am1	`optic`	1,193	0.56	99
LT33	`optic`	2,630	0.94	7
Li14	`optic`	1,316	0.96	14

Among the T4/T5 targets, only LLPC1 can be the figure-output sheet

A first figure-output sheet must be excitatory, retinotopic (a columnar population rather than a single wide-field cell), and project out of the optic lobe — which excludes most of the T4/T5 broadcast. Classifying every annotated target of the 912 reciprocal T4/T5 by neurotransmitter and class (Fig. S6; Table S1), 75% of the broadcast stays inside the optic lobe: 48% onto intrinsic feedback neurons (TmY, Tm, Y, C3, Mi, Tlp; recurrent processing, no central output) and 24% onto optic GABAergic interneurons (LPi, Li, LT, CT1; inhibitory, not an excitatory readout). A further 13% reaches wide-field tangential and centrifugal cells (HS, VCH, DCH, CT1) — only 62 cells, which collapse retinotopy. Just 7% reaches cholinergic columnar visual-projection populations, the only class with the right properties, and within that class LLPC1 is dominant: it receives 17,499 synapses, roughly 40-fold more than each of its direction-sibling sheets (LLPC2, LLPC3, LPC1, LPC2; 318–401 synapses), which are most plausibly the parallel back-to-front and vertical sheets that this front-to-back VCH-gated T4a channel does not drive. The only other strongly-driven cholinergic columnar projection neuron is LPLC2 (5,124 synapses), the canonical looming/expansion detector [44, 45] — a feature pathway whose jump/escape output is distinct from LLPC1's wing-steering readout. Other lobula visual-projection neurons that form optic glomeruli encode object and feature signals through parallel channels [46–50]. LLPC1 is therefore the unique candidate for the first retinotopic figure-output sheet of this branch.

A retinotopic LLPC1 sheet downstream of VCH-gated T4/T5

Having identified LLPC1 as the only viable figure-output sheet, we now characterize how it reads the VCH-gated motion.

We start from the 454 right-hemisphere T4a neurons that are postsynaptic to left VCH. These VCH-targeted T4a neurons make 9,223 synapses onto 100 LLPC1 neurons, all in the right optic lobe. Each LLPC1 pools about 15 VCH-gated T4a drivers, and each T4a contacts only a few LLPC1 cells. The anatomical scale is therefore intermediate: local enough to preserve position, but broad enough to pool over a small motion patch. The full input census shows that T4a and T5a dominate the sheet's excitatory synaptic mass, whereas the most common shared inputs are broad inhibitory or centrifugal cells, including LPi15, PVLP011, Am1, VCH and DCH. This defines a local excitatory sheet embedded in shared inhibition.

This pooling is statistically local, not merely convergent. In a null model in which each LLPC1 samples the same number of T4a inputs at random from the VCH-targeted pool (preserving in-degree), the expected input-patch radius equals the whole field ($\sim 40\,\mu$m); the observed median is $6.3\,\mu$m, below all 500 permutations ($z = -68$; $p < 0.002$), and 92% of each cell's T4a input falls within $10\,\mu$m of its centroid versus 10% under the null (Fig. S1). The convergence is therefore a spatially ordered map — the anatomical substrate required for a position-coded figure signal — rather than random or merely dense sampling.

LLPC1 also reads lobula form input: a motion–form integration layer

LLPC1 is named for the two neuropils it spans — the lobula and the lobula plate — and its input wiring shows that it integrates the distinct visual streams they carry. Beyond the lobula-plate motion channel, LLPC1 receives a comparably large input from *lobula* neurons carrying non-direction-selective form and object information: transmedullary Tm and TmY cells together with Y, Tlp and Li neurons (Fig. 4a). These lobula-projecting medulla and lobula neurons

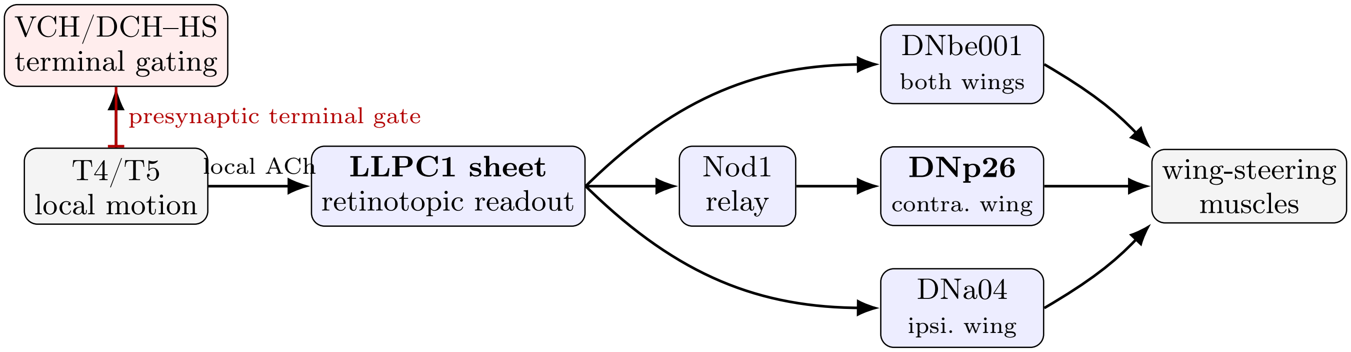

Figure 1: Proposed circuit. *Stage 1 (separation):* VCH pools wide-field T4/T5 motion and feeds GABA back onto the T4/T5 terminals, where it is positioned to suppress common-mode motion so that the signal reaching LLPC1 is largely background-gated. *Stage 2 (readout and steering):* the cholinergic LLPC1 sheet reads out the gated local signal retinotopically and drives wing-steering descending neurons — the Nod1 relay to the steering command neuron DNp26, and the direct targets DNbe001 and DNa04. In the male whole-CNS connectome these neurons project to the direct wing-steering muscles, with DNp26 biasing the contralateral wing, DNa04 the ipsilateral wing and DNbe001 both. Sheet-regulating inhibition (LPi15, PVLP011) and the weaker PLP/PVLP broadcast route to escape are omitted for clarity.

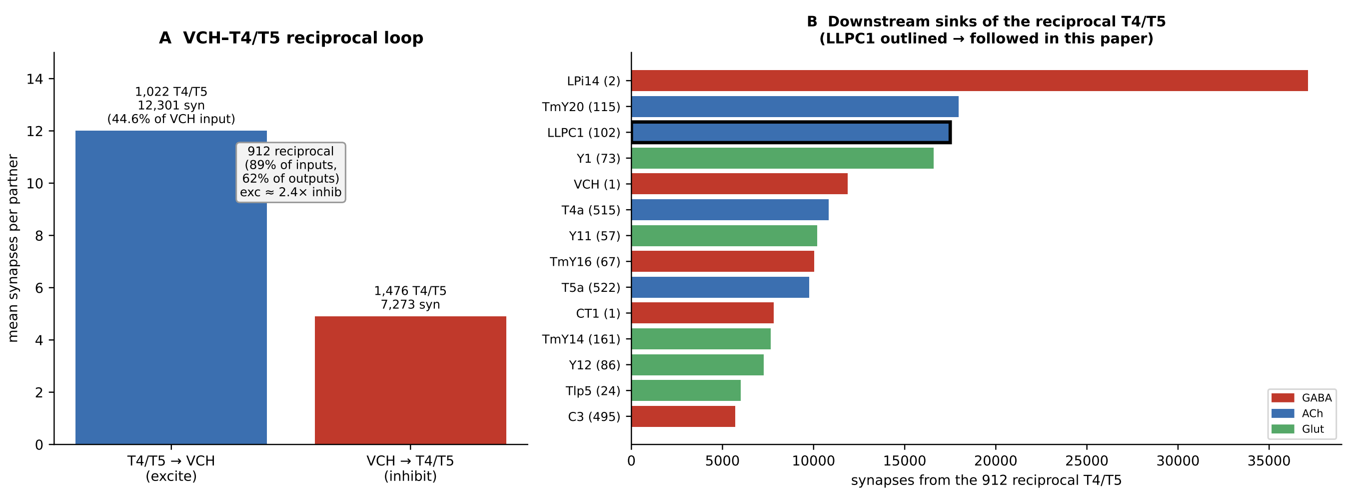

Figure 2: The VCH–T4/T5 reciprocal loop is the circuit's entry point. a, Left VCH and its T4/T5 partners: 1,022 T4/T5 neurons drive VCH (12,301 synapses, 44.6% of VCH input; mean 12.0 syn each), VCH inhibits 1,476 T4/T5 (7,273 synapses; mean 4.9 syn each), and 912 are reciprocal (89% of inputs), giving a densely reciprocal gain-control loop in which excitation is ∼2.4× the inhibitory feedback. **b**, downstream sinks of the 912 reciprocal T4/T5 by synapse count, coloured by neurotransmitter: lobula-plate intrinsic (LPi14), TmY, the cholinergic LLPC1 population (outlined), VCH itself, CT1, HS and DCH. LLPC1 is the projection-neuron sheet followed in the rest of this paper.

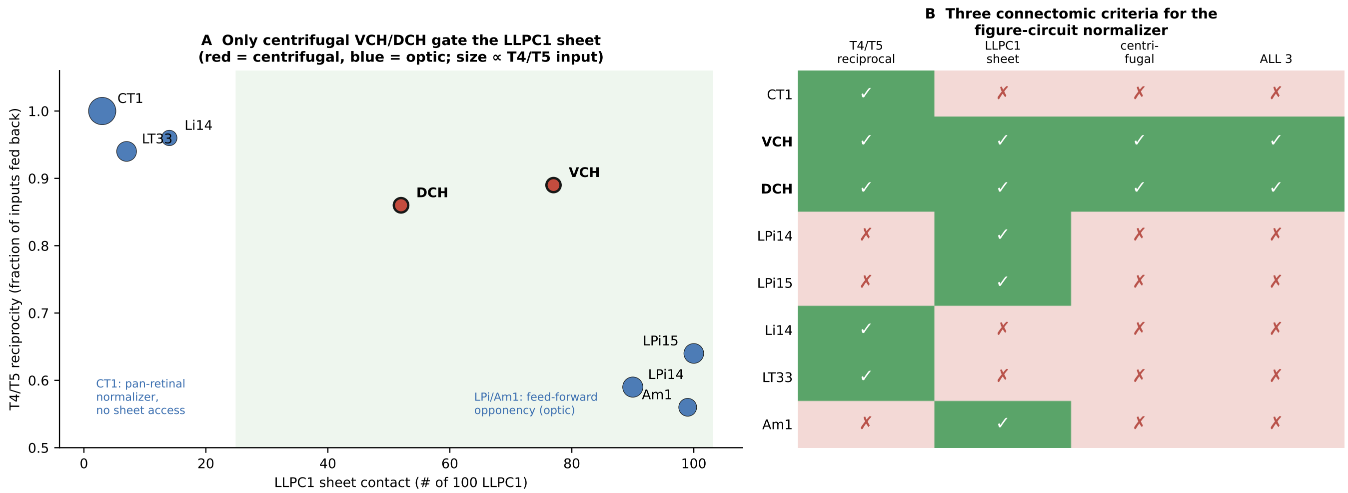

Figure 3: VCH is the only centrifugal, T4/T5-reciprocal inhibitor that gates the LLPC1 sheet. **a**, each large inhibitory T4/T5 target plotted by LLPC1-sheet contact (x) and T4/T5 reciprocity (y); marker colour is super-class (red, centrifugal; blue, optic) and size scales with T4/T5 input. CT1 is reciprocal but has no sheet access; LPi14/LPi15/Am1 contact the sheet but are optic and feed-forward; only VCH and DCH combine both with centrifugal identity. **b**, the same eight cells scored on the three criteria; only VCH and DCH satisfy all three.

carry contrast- and form-related signals rather than direction-selective motion [15, 22, 24]. Across the 100 LLPC1, direction-selective T4/T5 motion supplies 29% of all input synapses while lobula form/object neurons supply 31% — an even split between motion and form, the remainder being shared inhibition.

The two streams arrive on *separate dendritic fields*. Snapping every input synapse onto the LLPC1 skeletons, the T4/T5 (lobula-plate) and Tm/TmY (lobula) synapses occupy spatially distinct arbors: across all 100 cells the two field centroids are a median 20 μm apart, twice the radius of either field (Fig. 4b,c). Each LLPC1 therefore carries one dendritic tuft in the lobula plate, sampling local motion, and a second in the lobula, sampling form and object features, and combines them into a single columnar output.

This binding of motion and form is why an additional layer is required before action. Direction-selective T4/T5 motion alone cannot specify *what* the moving figure is, and lobula form channels alone cannot specify that it is moving differently from the background; only a neuron that reads both retinotopically — as LLPC1 does — can represent a motion-defined *object* for routing to steering descending neurons. LLPC1 is therefore not a passive relay of the VCH-gated motion residual but the convergence stage where motion-defined and feature-defined figure information are bound into one retinotopic output.

A two-stage circuit: VCH gates the terminals; LLPC1 reads out under sheet regulation

The connectome assigns the separation step to VCH, acting *upstream* of LLPC1. VCH receives 12,301 synapses from 1,022 T4/T5 neurons, dominated by balanced T4a and T5a, and sends 7,273 synapses back to 1,476 T4/T5 neurons; 912 of these are reciprocal, and almost all of them (882 of 912) belong to a single horizontal direction (layer-a, front-to-back; Fig. S7a), forming a densely reciprocal *same-direction* gain-control loop. Crucially, VCH's inhibitory output lands on the T4/T5 axon terminals that drive LLPC1 (below), positioning it to suppress coherent wide-field motion at the terminal, before the sheet is reached. VCH is therefore the minimal background separator of this branch, not itself a retinotopic output.

The two other large inhibitors act on the sheet itself and *regulate*, rather than create, the separated signal. LPi15 contacts all 100 LLPC1 cells with 6,472 inhibitory synapses and is itself driven almost entirely (94.8%) by T4/T5 of the *opposite* horizontal direction (layer-b, back-to-front), whereas VCH and LLPC1 read layer-a (Fig. S7a). LPi15 is therefore a sheet-wide, opponent-direction feed-forward inhibition that sets the operating point [32, 33]; whether it implements literal opponent subtraction or common-mode suppression is not resolved by the anatomy alone. PVLP011 receives from all 100 LLPC1 and feeds GABA back to 99 of them on a distal output compartment (below), making it a recurrent regulator of sheet gain and persistence rather than part of the first-pass separation. These roles are experimentally separable from VCH: silencing VCH should make LLPC1 responses more raw-motion-like and less

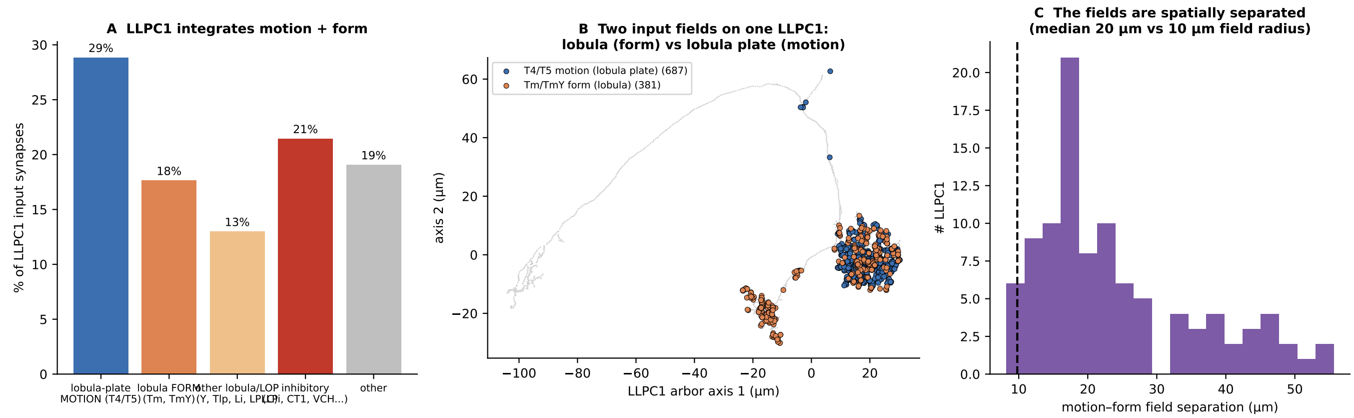

Figure 4: LLPC1 integrates lobula-plate motion with lobula form on two separate dendritic fields. a, input composition of the 100 LLPC1: direction-selective T4/T5 motion (lobula plate, 29%) and lobula form/object neurons (Tm, TmY, Y, Tlp, Li, LPLC; 31%) supply comparable excitatory input, alongside shared inhibition. **b**, one example LLPC1 skeleton: T4/T5 (motion) and Tm/TmY (form) input synapses cluster in two spatially distinct arbors. **c**, across all 100 LLPC1 the motion and form input fields are separated by a median 20 μm, twice the $\sim$10 μm field radius (dashed) — two dendritic fields, one in the lobula plate and one in the lobula.

figure-selective, whereas silencing LPi15 or PVLP011 should change the sheet's operating point, gain or persistence without abolishing the VCH-gated signal (Figs. S2, S3).

Synapse location confirms the two-stage architecture. Mapping each input onto the local FlyWire skeletons (cable distance; Fig. S4) shows that these operations are also spatially segregated. On the T4a side, VCH$\to$T4a synapses lie a median of only 2.3 μm (cable) from the T4a$\to$LLPC1 *output* terminal, versus 51 μm for the T4a's other inputs — VCH is the closer input in 282 of 286 T4a with skeletons ($p \approx 7 \times 10^{-48}$, Wilcoxon). VCH is therefore positioned to gate the specific T4a release sites that drive LLPC1, the anatomical signature of presynaptic/terminal gain control. On the LLPC1 side, LPi15 (1.3 μm) and VCH (2.0 μm) inhibitory synapses are interdigitated with T4a excitation on the dendrite, whereas PVLP011 synapses lie a median of 185 μm away on a distinct, distal compartment. This is the anatomical basis of the two stages: VCH suppresses motion at the T4/T5 *output terminal* (stage 1, presynaptic), whereas the sheet-regulating inhibitors act later — LPi15 on the LLPC1 dendrite (operating-point control) and PVLP011 on the LLPC1 output compartment (recurrent regulation).

LLPC1's output is dominated by a single excitatory readout, Nod1

Having shown how LLPC1 builds a retinotopic figure signal, we asked where that signal goes. The 100 LLPC1 cells make 106,269 output synapses, 39% of them onto central-brain, projection or descending neurons outside the optic lobe (Fig. S7c). Classifying every shared target by neurotransmitter and class (Fig. 5a) shows that the sheet contacts many cells but has a single dominant *excitatory* readout. Its two most heavily contacted targets are inhibitory and cannot forward an excitatory figure signal: PLP249 (4,389 synapses; 97 of 100 LLPC1; GABAergic) acts as a feed-forward selector, and PVLP011 (3,168 synapses; 100 of 100; GABAergic) is the recurrent gain-control cell that feeds back onto the sheet. Among cholinergic targets, the cell type Nod1 — two central-brain-projecting visual-projection neurons — dominates: it receives 4,228 synapses from 91 of the 100 LLPC1, about 1.7-fold more than the next excitatory readout PLP163 (2,534 synapses; 100 of 100) and roughly six-fold more than the strongest *direct* descending target DNbe001 (724 synapses). The remaining shared targets are recurrent or feedback elements — the lateral LLPC1 network, glutamatergic Y and Tlp medulla feedback, and LPi inhibition. Nod1 is therefore the principal excitatory forward channel out of the figure sheet.

Nod1 relays the figure signal to the steering command neuron DNp26

We then asked where the figure signal actually reaches motor control. Across all routes the sheet contacts 125 descending neurons of 89 types — through a small direct channel (2,174 synapses to 25 DNs), the Nod relay and a PLP/PVLP broadcast route — but its steering output is concentrated on Nod1. The two Nod1 cells make 662 synapses onto 27 descending neurons, and 97% of this descending output (644 of 662 synapses) targets known steering descending neurons

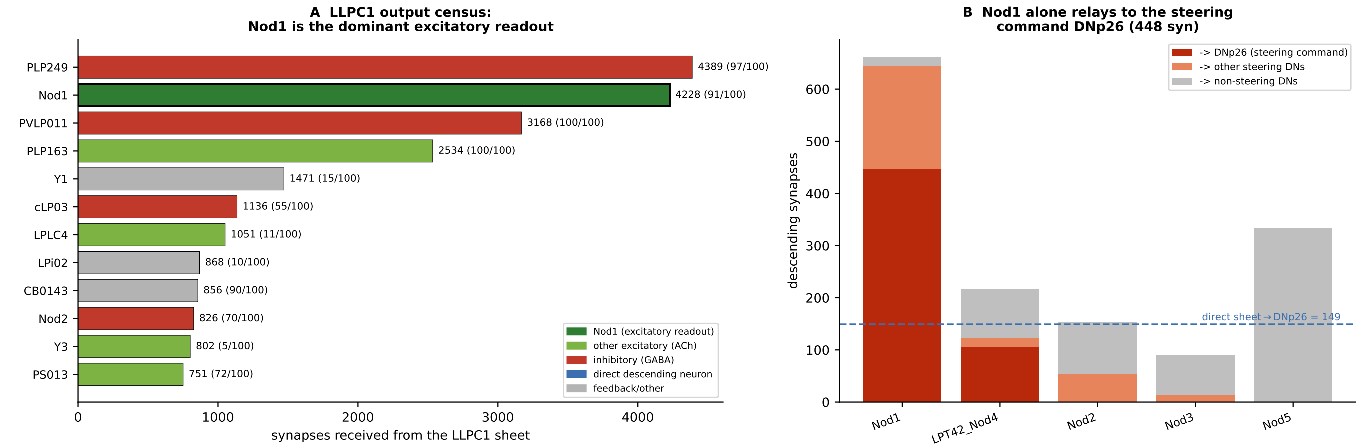

Figure 5: Nod1 is the dominant excitatory readout of LLPC1 and the relay to the steering command neuron DNp26. **a**, LLPC1 output census: synapses received from the sheet by each major shared target, coloured by role (number of the 100 LLPC1 contacting it in parentheses). The two largest targets, PLP249 and PVLP011, are GABAergic (feed-forward selector and recurrent gain control); among excitatory targets the cholinergic Nod1 dominates. **b**, descending output of each Nod-type cell, partitioned into the steering command neuron DNp26, other known steering DNs and non-steering DNs. Only Nod1 relays substantially to DNp26 (448 synapses), exceeding the direct sheet→DNp26 input (149; dashed line); Nod5 instead drives the neck/gaze DN DNb03, and Nod2 is GABAergic.

(Fig. 5b). It is overwhelmingly focused on one cell: DNp26 receives 448 synapses from Nod1 — three-fold more than the 149 synapses it receives *directly* from the sheet, and four-fold more than from any other Nod-type cell. DNp26 is the strongest convergent course-control target of the entire circuit, and the male whole-CNS connectome resolves it as wing-steering-biased, with prominent hg1, hg2 and i1 steering motor-neuron targets (Fig. S9). Nod1 is thus the principal relay that converts the LLPC1 figure-motion signal into a steering command.

What singles Nod1 out is that the other heavily shared LLPC1 targets do not carry this steering signal. Nod2 is GABAergic and inhibitory; Nod5 sends its descending output to the neck/gaze neuron DNb03 (247 synapses) rather than to wing steering; Nod3 is weak; and the GABAergic PLP249 and PVLP011 regulate the sheet rather than forwarding it. The cholinergic PLP163 and the remaining PLP/PVLP cells form a separate, weaker broadcast route whose distinctive feature is that it is the *only* route to the looming/escape command cluster, including the giant fibre DNp01, DNp03, DNp04 and DNp06 [44, 45, 51, 52]. The whole-CNS motor map confirms this division of labour: the figure-carrying direct and Nod-type channels terminate on wing-steering, haltere and neck/gaze motor systems (the Nod-type arm is 55% wing-steering and 14% neck/gaze, with 1.6% jump), whereas the broadcast route is the only channel with substantial jump/TTM output (17%; Fig. S9). The figure-motion sheet therefore has a dedicated steering output through Nod1 and DNp26, kept anatomically separate from the escape pathway.

The descending targets are wing-steering neurons, and several control a specific wing

Finally, we characterized the descending neurons that carry the figure signal to the wings. Seven DN types receive substantial input from the sheet or from Nod1 and have wing steering as their dominant motor system in the male whole-CNS connectome (Fig. 6a): the directly driven DNbe001 (1,181 wing-steering motor-neuron synapses) and DNa04 (908), and the Nod1-driven DNp26, DNg32, DNge094, DNge107 and DNbe005. These belong to, or connect with, descending and central classes implicated in flight and walking steering, visual saccades, steering-command formation and state-dependent gating [53–60]. Their output converges on the direct steering muscles — the basalar (b1–b3), first-axillary/pterale (i1, i2) and hg1–hg4 muscles that set wing-stroke amplitude and angle — and not on the bilateral power muscles (the dorsal-longitudinal and dorsoventral muscles, which we excluded as wing-power). The circuit's descending output is therefore directed at the steering, not the power, musculature.

Because the male connectome labels each motor neuron by the body side it innervates, we could ask whether a given DN controls a specific wing. Partitioning each DN's wing-steering output into the ipsilateral and contralateral wing reveals a clear division (Fig. 6b). DNa04 drives the ipsilateral wing almost exclusively (99% of its steering synapses). DNp26 — the principal Nod1 target — and DNg32, both driven through the Nod1 relay, instead drive the *contralateral* wing (77% and 97% of their steering output), as does DNge094 (100%). DNbe001, the strongest direct target, drives

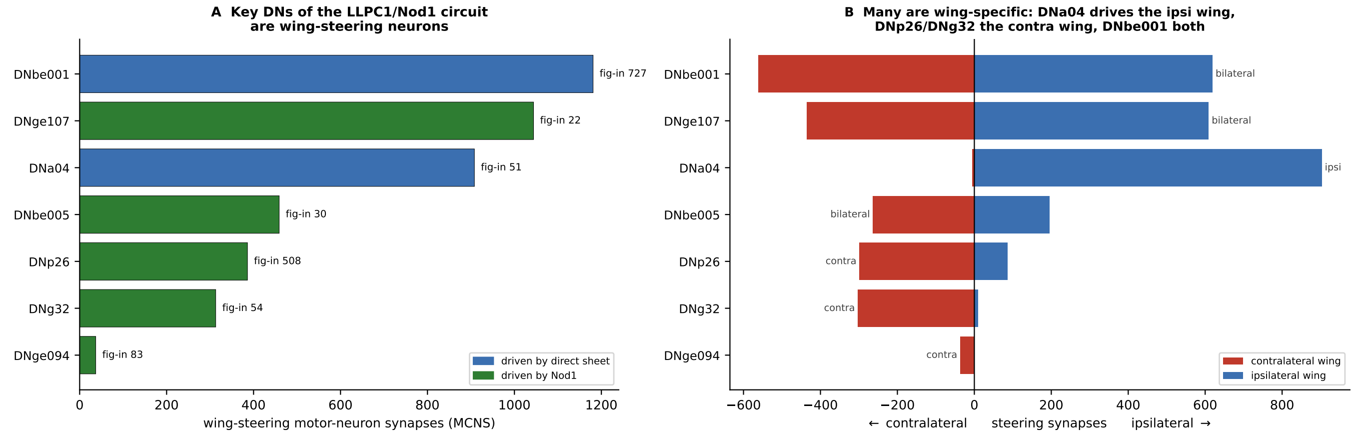

Figure 6: The descending targets of the LLPC1/Nod1 circuit are wing-steering neurons, several of which control a specific wing. **a**, the seven figure-driven DN types with wing steering as their dominant motor system, by wing-steering motor-neuron synapses in MCNS, coloured by whether they are driven by the direct channel (blue) or the Nod1 relay (green); their figure input (sheet+Nod1 synapses) is annotated. **b**, each DN's wing-steering output partitioned into the ipsilateral (blue) and contralateral (red) wing. DNa04 drives the ipsilateral wing; the Nod1-driven DNp26, DNg32 and DNge094 drive the contralateral wing; DNbe001 and DNge107 are bilateral.

both wings symmetrically (52% ipsilateral). The figure circuit thus issues both bilateral commands (DNbe001) and lateralized, single-wing commands — ipsilateral through DNa04 and contralateral through DNp26 and DNg32 — the anatomical substrate for the asymmetric left-versus-right wing adjustments that produce a turn.

This wing specificity is concrete at the level of individual muscles. DNp26, the convergent steering command at the end of the LLPC1→Nod1 pathway, makes its strongest *contralateral* connections onto the hg1 (122 synapses) and i1 (105) steering motor neurons, together with b3 (28), the muscles that modulate wing-stroke amplitude and angle during turning. The pathway from VCH-gated T4/T5, through the retinotopic LLPC1 sheet and the cholinergic Nod1 relay, therefore terminates on identified steering muscles of one wing — an unbroken anatomical route from a motion-defined figure to a wing-specific steering command.

Discussion

Together, these analyses define a two-stage figure-ground circuit with a specific motor output. Upstream, VCH performs the background separation by presynaptically gating the T4/T5 terminals that drive LLPC1; LLPC1 is then the first retinotopic sheet that reads out this gated signal, under sheet-wide opponent-direction LPi15 inhibition and recurrent PVLP011 regulation. Downstream, LLPC1 does not merely feed a central broadcast route: it routes figure-motion residuals mainly through direct and Nod-type descending channels whose motor footprints are wing-steering, haltere, neck/gaze, abdominal and wing-power systems, and it reaches escape musculature only through the weaker PLP/PVLP broadcast route.

This has immediate experimental consequences. Nod1 and its principal target DNp26 should be the first elements to characterize: Nod1 is the dominant excitatory readout of the sheet (4,228 synapses from 91 of 100 LLPC1) and routes 97% of its descending output to steering DNs, while DNp26 is its strongest target and the convergent wing-steering command neuron. DNbe001, the strongest *direct* descending target (70 of 100 LLPC1; resolved by MCNS as a wing-steering/flight-control DN), is a second priority. Because DNp26 drives the contralateral wing's hg1/i1 steering muscles whereas DNa04 drives the ipsilateral wing, activating these neurons should bias steering toward opposite sides — a directly testable consequence of the wing-specific wiring. Imaging LLPC1, Nod1 and DNp26 during figure-with-moving-ground assays should reveal whether these outputs encode local motion residuals rather than raw wide-field motion. Perturbing VCH should remove the terminal gate and make LLPC1 more raw-motion-like, whereas perturbing LPi15 or PVLP011 should change the sheet's operating point or recurrent gain without abolishing the gated signal. Finally, LPLC2/LC4 escape stimuli should remain separable: direct and Nod-type LLPC1 routes should bias steering, whereas jump/TTM output should require the broadcast route or parallel looming circuits.

The model is a circuit proposal, not a claim that connectome weights alone determine physiology. Electrical coupling, synaptic strength, state dependence, neurotransmitter dynamics, and sex differences between female FAFB and male MCNS remain open. Nevertheless, the anatomical logic is unusually convergent: the same sheet that receives local VCH-gated motion and shared inhibition also reaches wing-steering motor systems through focused descending routes. That makes LLPC1 the strongest current candidate for the retinotopic visual representation that converts motion-defined figure-ground separation into course-control behavior.

Methods

All optic-lobe, LLPC1 and DN-routing claims are connectomic analyses of FlyWire FAFB materialization v783, using chemical synapses and Schlegel et al. cell-type annotations. The downstream motor readout uses MCNS (`male-cns:v1.0`) because it contains both brain and ventral nerve cord, allowing FlyWire DN names to be matched to monosynaptic motor-neuron targets. Synapse counts are used as anatomical evidence, not as direct measures of synaptic efficacy. Full methods — synapse-query parameters, cell-type and neurotransmitter assignment, skeletonization and cable-distance computation, the retinotopy null model and statistics, and the motor-target classification — are given in the Supplementary Information, together with the complete decision ledger, per-channel descending-neuron tables and alternative-model evaluation.

Supplementary Information

Extensive appendix for the LLPC1 course-control manuscript

Overview. This Supplementary Information gives the full methods, statistics and per-analysis evidence behind the main text: the connectomic datasets and analysis pipeline (Methods); the directional, pooling and projection evidence that identifies LLPC1 as a first retinotopic figure-output sheet; the layered inhibitory architecture and its cable-distance compartmentalization; the VCH–T4/T5 normalization loop; the descending-channel and whole-CNS motor-target tables; a full evaluation of alternative circuit models; the computational formulation; and discriminating experimental predictions, limitations and reproducibility.

S1 Methods

Connectomes and versions. All optic-lobe, LLPC1 and descending-neuron analyses use the adult female FlyWire FAFB connectome [34] at materialization v783, with chemical synapses from the `synapses_nt_v1` table and hierarchical cell-type and super-class annotations from the whole-brain annotation framework [35] and the optic-lobe parts list [36,61]. The motor read-out uses the male CNS connectome (MCNS, `male-cns:v1.0`), which contains both the brain and the ventral nerve cord and therefore links descending-neuron cell types to their motor-neuron targets.

Synapse queries. Connectivity was queried from the CAVE materialization service. Unless stated otherwise we use all chemical synapses with no cleft-score threshold, which reproduces the published synapse totals; robustness of the principal sheet claims to cleft-score and per-edge thresholds is reported in the Statistical robustness section below. Counts onto unannotated segments are reported as partner-segment counts, not neuron counts, because many are small unproofread fragments.

Cell-type and neurotransmitter assignment. Cell types, super-classes and sides were taken from the FlyWire annotation tables [35, 36]. Predicted neurotransmitter is used as an anatomical annotation, not as an established transmitter identity. In particular, physiological studies show that the opponent inhibition supplied by lobula-plate intrinsic cells is glutamatergic and acts through glutamate-gated chloride channels [32, 33], whereas the connectome predictor labels some LPi types GABAergic; we therefore treat LPi15 as inhibitory of uncertain transmitter. T4/T5 are treated as cholinergic excitatory neurons, consistent with their excitatory transmission onto lobula-plate cells [32].

Directional subtypes. T4 and T5 each comprise four subtypes (a–d) whose axons terminate in the four layers of the lobula plate and which are tuned to the four cardinal directions of motion [15,17,25]. We use the convention that layer-a (layer 1) corresponds to front-to-back and layer-b (layer 2) to back-to-front horizontal motion. Directional composition was computed by classifying each T4/T5 partner by its annotated subtype and summing synapses.

Skeletons and cable distance. Synapse-location analyses used the FlyWire skeleton node CSVs (the summary status flags are stale, but the skeletons carry full morphology, $\sim$0.9–2.3 mm cable). Each synapse was snapped to the nearest skeleton node and geodesic (along-cable) distances computed with `navis`. Usable skeletons exist for 286/454 T4a and 88–100/100 LLPC1; conclusions are sign-identical under a Euclidean metric.

Retinotopy null model. Pooling locality was tested against an in-degree-preserving null in which each LLPC1 draws its observed number of distinct T4a inputs uniformly at random from the VCH-targeted pool, recomputing the synapse-weighted input-patch radius over 500 permutations; significance is reported as a z-score against the null distribution and a permutation p-value.

Input-field and statistics. For the motion/form field separation, each input synapse was assigned to its presynaptic cell type and snapped onto the post-synaptic LLPC1 skeleton; field centroids and radii were computed per cell. Paired synapse-location comparisons used the two-sided Wilcoxon signed-rank test across neurons. Population summaries are medians with interquartile ranges unless noted.

Motor-target classification. Downstream DN cell types were matched into MCNS and their monosynaptic motor-neuron targets classified by curated motor-neuron subclass (jump/TTM, wing-steering, wing-power, leg, neck/gaze, haltere, abdominal). Channel percentages are circuit-weighted over the DN targets with resolvable monosynaptic motor connectivity. For the wing-specificity analysis, each wing motor neuron's innervated side was read from its `somaSide` annotation, and a DN's steering output was partitioned into the ipsilateral and contralateral wing using only the direct steering muscles (the DLM/DVM power muscles were excluded).

Laterality and scope. The analyzed sheet is the right-hemisphere LLPC1 reached by right-side T4a postsynaptic to left VCH; it is the front-to-back exemplar of a bilateral, multi-direction system.

Software and availability. Analyses used `caveclient`, `navis`, `numpy`, `scipy`, `pandas` and `matplotlib` on Python 3.9. FlyWire (`flywire_fafb_public`) and the MCNS connectome are public resources; analysis code and intermediate tables are available from the authors.

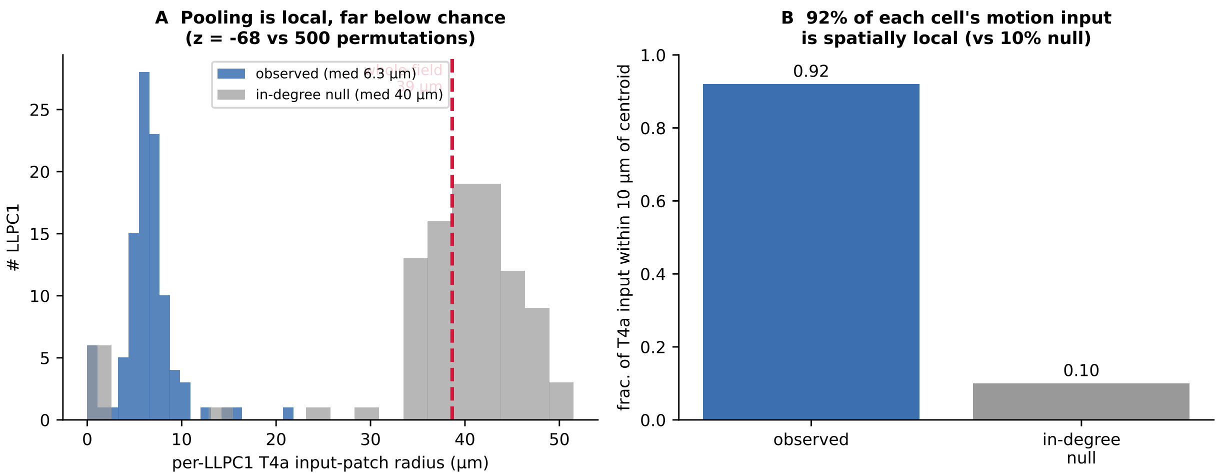

Figure S1: T4a→LLPC1 pooling is retinotopically local, far below chance. a, observed per-LLPC1 input-patch radius (median 6.3 μm) versus an in-degree-preserving null (median $\sim$40 μm $\approx$ the whole field); the distributions do not overlap ($z = -68$ across 500 permutations). **b**, 92% of each LLPC1's T4a input lies within 10 μm of its centroid, versus 10% expected under the null. The retinotopic convergence is therefore a statistical property of the sheet rather than an artefact of dense or random sampling.

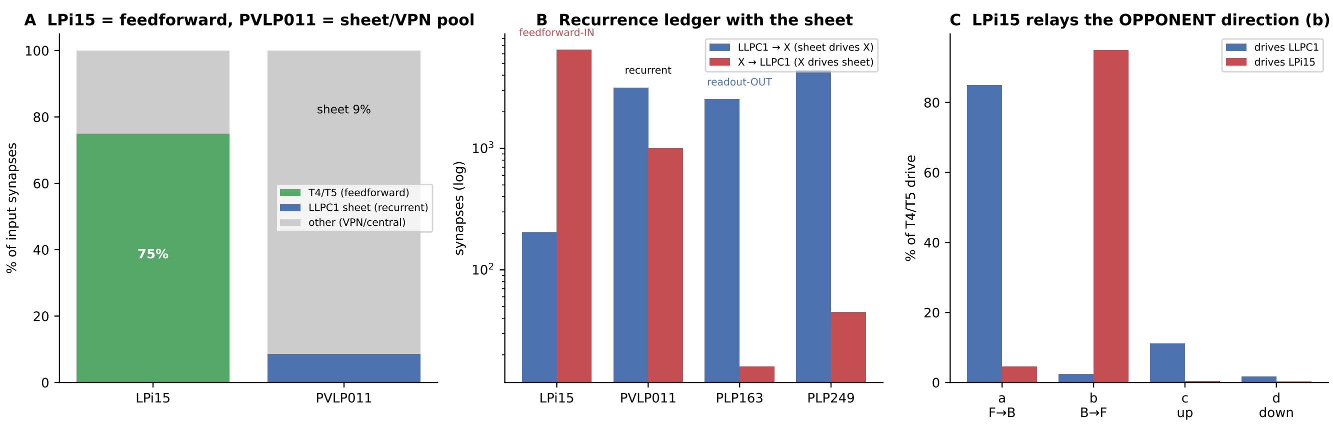

Figure S2: Inhibition is decomposed into feed-forward, wide-field and recurrent operations. LPi15 is driven primarily by T4/T5 and acts as a universal GABAergic sheet input; PVLP011 is the principal recurrent GABAergic readout/input loop; VCH/DCH–HS supplies wide-field context and first-stage T4/T5 gain control. The inhibitory input to the sheet thus comprises separable feed-forward, wide-field and recurrent operations rather than a single undifferentiated pool.

S2 Extended data figures

Table S1: The 912 reciprocal T4/T5 broadcast onto annotated targets, by functional class. Only the cholinergic columnar visual-projection sheets (LLPC/LPC) have the excitatory, retinotopic and central-projecting properties of a figure-output sheet.

Target class	% broadcast	cells	verdict
Optic intrinsic feedback (TmY/Tm/Y/C3/Mi/Tlp)	48	10,078	recurrent; stays in optic lobe
Optic inhibitory (LPi/Li/LT/CT1)	24	1,646	GABAergic; not an excitatory readout
Wide-field tangential/centrifugal (HS/VCH/DCH)	13	62	collapse retinotopy
Columnar projection sheet (LLPC/LPC)	**7**	315	**excitatory, retinotopic, projecting**
Other visual-projection (LPT/Nod)	5	279	relays, not the first sheet
Looming/object VPN (LPLC/LC)	2	224	feature/escape pathway

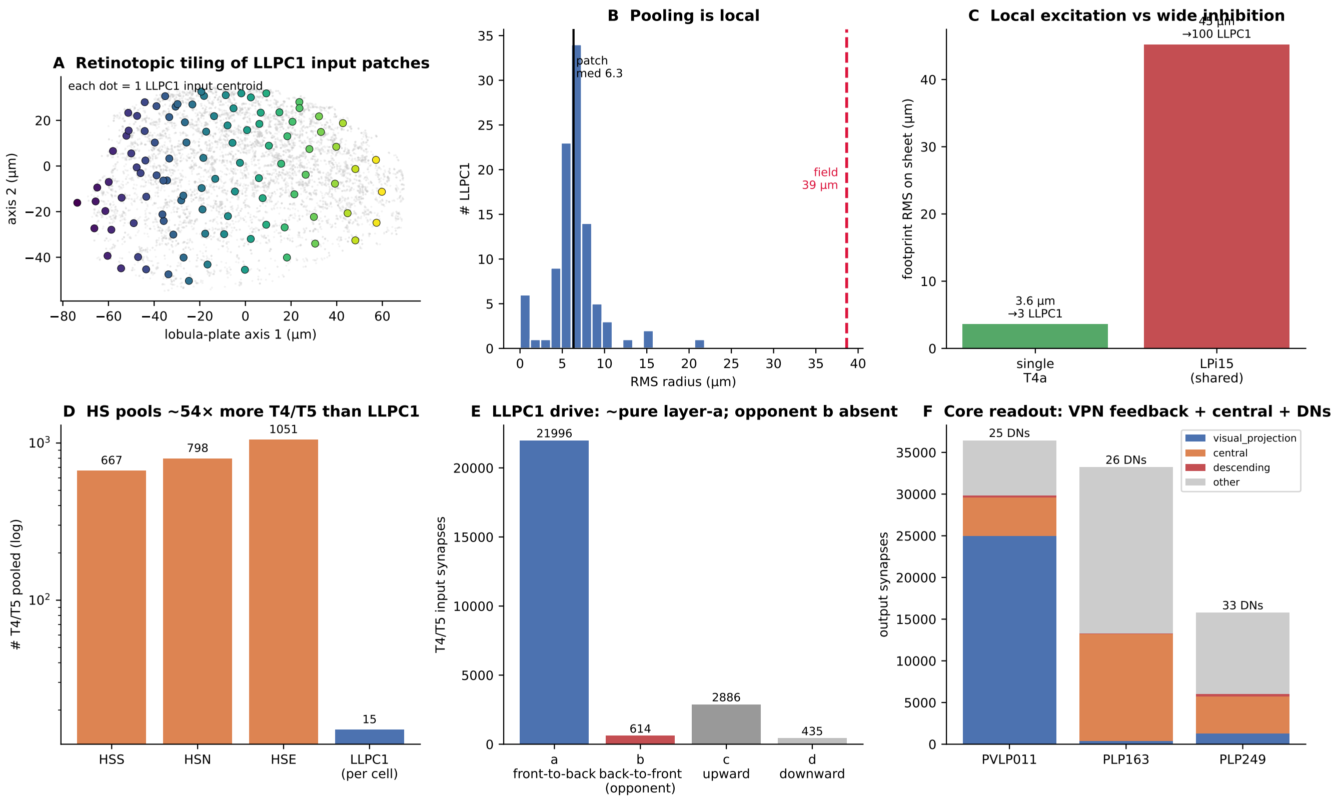

Figure S3: Connectomic support for a local excitatory sheet under wide-field inhibition. LLPC1 shows compact T4a→LLPC1 pooling, a small per-cell input patch, broad sheet-wide inhibition, and a marked contrast between single T4a footprints and the universal LPi15 footprint. HS-family cells pool far more T4/T5 inputs per cell than LLPC1, consistent with a wide-field optomotor role rather than a first retinotopic figure sheet.

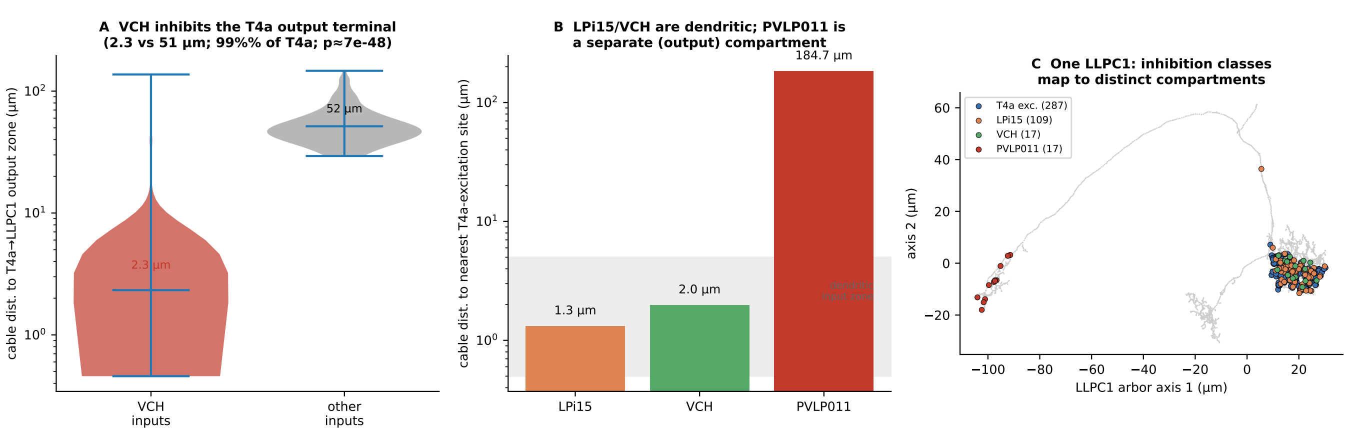

Figure S4: Cable-distance synapse-location analysis confirms compartmentalized inhibition. **a**, On the T4a, VCH input synapses sit a median 2.3 μm (cable) from the T4a→LLPC1 output terminal versus 51 μm for other inputs (VCH closer in 99% of 286 T4a; $p \approx 7 \times 10^{-48}$) — the signature of presynaptic gain control. **b**, On the LLPC1, LPi15 (1.3 μm) and VCH (2.0 μm) inhibition are interdigitated with T4a excitation (dendritic "input zone," grey band), whereas PVLP011 lies 185 μm away. **c**, one example LLPC1 skeleton: T4a, LPi15 and VCH synapses cluster in the dendritic tuft while PVLP011 occupies a separate distal branch.

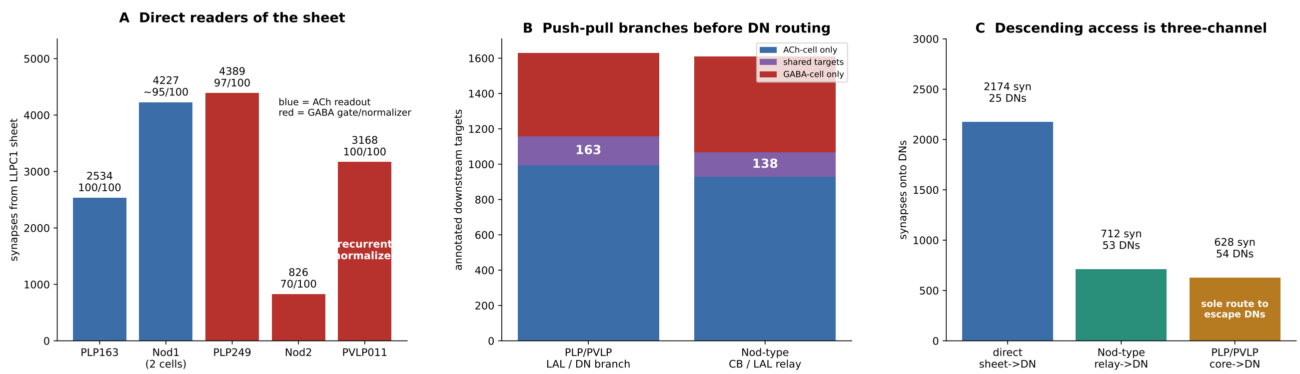

Figure S5: The shared PLP/PVLP readout group is not one functional cell. PLP163, PLP249 and PVLP011 are all shared LLPC1 targets, but their neurotransmitter predictions and recurrence differ. PLP163 is the strongest candidate excitatory readout, PLP249 is a GABAergic selector, and PVLP011 is a recurrent output-gain controller whose synapses lie on a distal LLPC1 compartment (Fig. S4), feeding back to nearly the entire sheet.

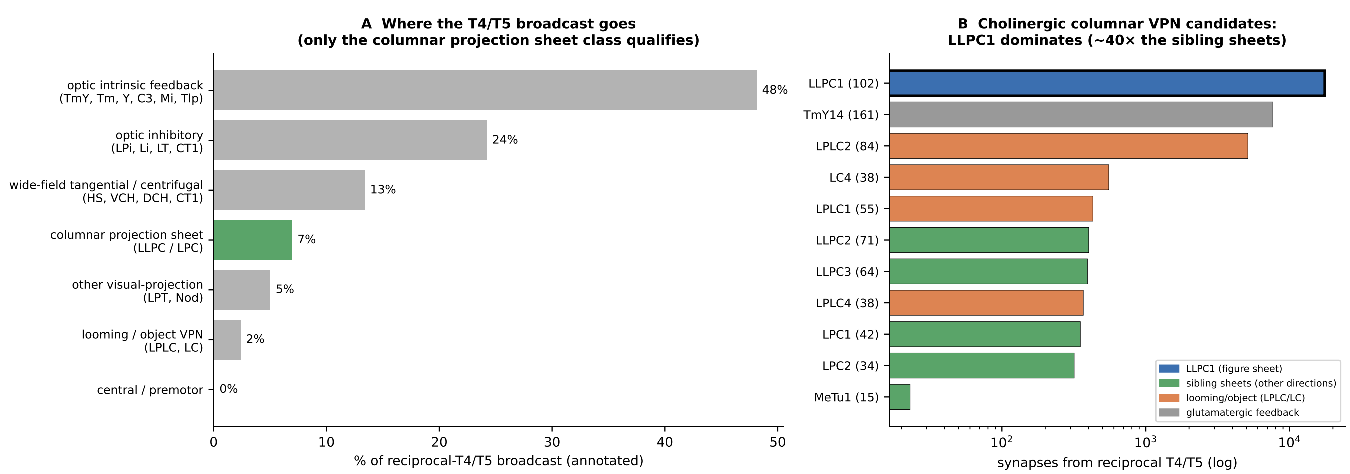

Figure S6: Candidate elimination among the T4/T5 downstream targets. **a**, the 912 reciprocal T4/T5 broadcast (annotated targets) by functional class: 48% to optic-intrinsic feedback, 24% to optic GABAergic interneurons, 13% to wide-field tangential/centrifugal cells, and only 7% to cholinergic columnar projection sheets (LLPC/LPC, green) — the only class that can be an excitatory, retinotopic, central-projecting figure output. **b**, within the cholinergic columnar visual-projection candidates (log scale), LLPC1 receives ∼40× more T4/T5 input than its direction-sibling sheets (LLPC2/3, LPC1/2); LPLC2/LC are the looming/object pathway and TmY14 is glutamatergic feedback.

S3 Datasets, nomenclature and analysis scope

Connectomes and versions. The optic-lobe and brain wiring claims in the main text are based on FlyWire FAFB materialization v783 with chemical synapses from the `synapses_nt_v1` table and annotations from the Schlegel et al. whole-brain annotation framework. The VCH–T4/T5 analysis, LLPC1 microcircuit analysis and DN enumeration were all derived from this FlyWire materialization. The motor-output analysis uses the male CNS connectome (MCNS; `male-cns:v1.0`), because it contains both the brain and the ventral nerve cord, allowing FlyWire DN cell-type names to be matched to motor-neuron targets. We use MCNS rather than the ventral-nerve-cord-only MANC connectome because MANC does not resolve many of the connectome-only "e"-name DN types, including DNbe001, DNae002 and DNpe056.

Laterality. The analyzed sheet is the right-hemisphere LLPC1 sheet reached by right-side T4a neurons that are postsynaptic to left VCH. Left-hemisphere LLPC1 cells and other LLPC/LPC subtypes may implement homologous or direction-specific variants. We therefore treat the present circuit as the front-to-back right-side exemplar, not as the complete bilateral figure-motion system.

Terminology. We use "Nod-type" rather than "nodulus" for the LLPC1→Nod1/2/3/5 arm because these visual-projection neurons are unnamed, the one localized example arborizes in the LAL rather than clearly in the nodulus, and their downstream targets are central-brain/LAL dominated with only minor central-complex targets. The branch is therefore best described as a Nod-type central-brain/LAL relay pending direct neuropil confirmation; its central targets include regions where visual-motion responses are behaviourally gated [62, 63].

S4 Quantitative support for the LLPC1 sheet

Table S2: LLPC1 sheet input–output metrics.

Metric	Value
VCH-targeted T4a set	454 T4a receiving 3,445 VCH synapses
T4a→LLPC1 synapses	9,223
LLPC1 reached	100 of 219 brain-wide; all right optic lobe
T4a drivers per LLPC1	mean 14.8; median 15; max 32
LLPC1 with ≥10 / ≥20 T4a drivers	79 / 22
T4a fan-out to LLPC1	mean 3.3; median 3; max 8
LLPC1 total output synapses	106,269
Output onto targets shared by ≥50 LLPC1	22.4% of LLPC1 output
Central cells receiving from all 100 LLPC1	PVLP011, PLP163
Universal LLPC1 input	LPi15, 6,472 GABA synapses to all 100 LLPC1
VCH direct input to sheet	588 GABA synapses to 77/100 LLPC1

Table S3: Top shared LLPC1 outputs. "Drivers" denotes the number of LLPC1 cells contacting the target.

Target	Drivers/100	Syn	NT	Interpretation
PVLP011	100	3,168	GABA	recurrent output gain control (distal compartment); also common input
PLP163	100	2,534	ACh	excitatory PLP readout
PLP249	97	4,389	GABA	GABAergic selector; largest readout-group synapse count
Nod1	91	1,902	ACh	Nod-type central-brain/LAL relay
Nod1 copy 2	90	2,325	ACh	Nod-type central-brain/LAL relay
Nod2	70	826	GABA	inhibitory Nod-type relay
DNbe001	70	736	ACh	strongest direct descending target; MCNS wing-steering
DNae002	49	533	ACh	direct descending target; MCNS haltere-biased
LPi15	71	204	GABA	recurrent link, but mainly universal input
VCH	51	111	GABA	closes context loop

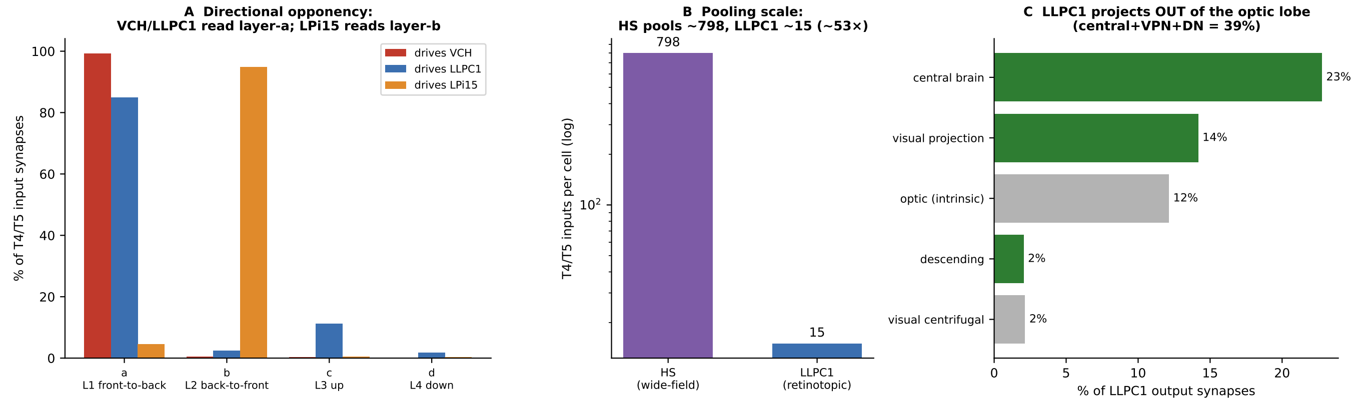

Figure S7: Directional opponency, pooling scale and central projection of the LLPC1 circuit. **a**, directional-subtype composition of the T4/T5 input to VCH, to the 100 LLPC1, and to LPi15: VCH (99.1%) and LLPC1 (84.8%) read layer-a (front-to-back), whereas LPi15 reads layer-b (back-to-front, 94.8%), the opponent horizontal direction. **b**, T4/T5 inputs per cell: HS tangential cells pool a median of ∼798 (wide-field), LLPC1 ∼15 (retinotopic); log scale. **c**, LLPC1 output by post-synaptic super-class; 39% leaves the optic lobe (central brain + visual projection + descending).

S5 Directional tuning, pooling scale and central projection

Three further measurements distinguish LLPC1 from the wide-field optomotor pathway and confirm the directional logic of the circuit (Fig. S7).

A same-direction reciprocal loop and opponent-direction sheet inhibition. Classifying every T4/T5 partner by directional subtype, the VCH loop and the LLPC1 sheet read a single horizontal direction whereas LPi15 reads the opposite one. Of VCH's 12,301 T4/T5 input synapses, 99.1% come from layer-a (front-to-back) T4a/T5a, and 882 of the 912 reciprocal partners are layer-a. LLPC1's direct T4/T5 drive is predominantly layer-a (84.8%), with a secondary upward (layer-c) component (11.1%) and little layer-b or layer-d. By contrast LPi15 — a single wide-field lobula-plate intrinsic cell that contacts all 100 LLPC1 — receives 94.8% of its 59,415 T4/T5 input synapses from layer-b (back-to-front) T4b/T5b. The sheet is therefore excited by one horizontal direction and inhibited, sheet-wide, by the opponent direction — the connectomic signature of motion opponency [32, 33].

Pooling scale. LLPC1 pools ∼15 T4/T5 per cell. The three horizontal-system (HS) tangential cells of the same hemisphere each pool a median of 798 T4/T5 inputs (mean 839; range 667–1,051), ∼50-fold more (Fig. S7b). HS therefore integrates almost the entire horizontal field, consistent with a wide-field optomotor role [26–29, 64], whereas LLPC1 samples a local patch — the difference between a wide-field stabilization cell and a retinotopic figure sheet.

Central projection. 39% of LLPC1 output synapses leave the optic lobe: 22.8% onto central-brain neurons, 14.2% onto other visual-projection neurons and 2.0% directly onto descending neurons, with only 12.1% returning to optic-lobe-intrinsic cells (Fig. S7c). A first figure-output sheet must project centrally, and LLPC1 does.

Table S4: Leading lobula (form/object) inputs to the 100 LLPC1 cells, by synapse count. These transmedullary (Tm), transmedullary-Y (TmY), Y, translobula-plate (Tlp) and lobula-intrinsic neurons project into the lobula and carry contrast- and form-related signals rather than direction-selective motion [15, 22, 24]. Together they supply 31% of LLPC1 input synapses, comparable to the 29% from direction-selective T4/T5.

Input type	Synapses to LLPC1	Class
TmY3	2,471	transmedullary Y (medulla–lobula–lobula plate)
TmY15	2,376	transmedullary Y
Y3	1,959	Y cell
Tm4	1,728	transmedullary (medulla→lobula)
TmY31	1,644	transmedullary Y
Tm3	1,555	transmedullary (medulla→lobula)
Y11	1,455	Y cell
Tlp4	1,334	translobula plate
Tlp1	1,321	translobula plate

S6 Statistical robustness and output completeness

Retinotopy null model. The locality of T4a→LLPC1 pooling was tested against an in-degree-preserving null in which each LLPC1 draws its observed number of distinct T4a inputs uniformly at random from the VCH-targeted pool, recomputing the synapse-weighted input-patch radius (500 permutations). The observed median patch radius ($6.3\,\mu$m) is ∼6.4-fold tighter than the null median ($40.0 \pm 0.5\,\mu$m) and lies below every permutation ($z = -67.7$; $p < 1/500$). The fraction of each cell's T4a input within $10\,\mu$m of its centroid is 0.92 observed versus 0.10 under the null (Fig. S1). This is the non-circular test of retinotopy; the per-synapse pre/post co-location used elsewhere is not relied upon.

Threshold robustness. The principal sheet claims are stable across edge- and cleft-score thresholds.

Table S5: Robustness of the core sheet claims to synapse thresholds (FAFB v783). Values are counts out of the 100-cell sheet.

Threshold	T4a→LLPC1 reach	LPi15→LLPC1	PVLP011 recurrence (in/out)
all synapses	100/100	100/100	100/99
cleft-score ≥50	95/100	100/100	100/99
edge ≥5 synapses	90/100	100/100	100/88
edge ≥10 synapses	75/100	100/100	95/50

LPi15's universality is threshold-invariant; PVLP011 recurrence and sheet membership survive cleft ≥50 and edge ≥5, thinning only when restricted to the strongest edges, as expected for a graded sheet.

Unannotated-output audit. Annotated neurons carry 53.1% of LLPC1 output synapses; the remaining 46.9% is onto unannotated segments. Thirty-nine such fragments are each contacted by ≥10 LLPC1; the most-shared is reached by 48/100. Because a single highly shared unannotated relay could revise the output model, these are flagged for proofreading.

Table S6: Top unannotated LLPC1 output fragments (reached by ≥10 LLPC1), by driver count.

Fragment root id	LLPC1 drivers	Syn	Status / action
720575940637089343	48	242	unannotated; top shared — prioritize proofreading
720575940627763243	35	62	unannotated
720575940625865697	24	61	unannotated
720575940626513889	23	66	unannotated
720575940608489225	18	30	unannotated

S7 Inhibitory architecture of the sheet

The quantitative input table below provides the numerical basis for the main-text decomposition of inhibition into LPi15 feed-forward opponency, VCH/DCH–HS wide-field context and PVLP011 recurrent output gain control. Transmitter labels follow the FlyWire prediction; physiological work shows that the opponent inhibition supplied by lobula-plate intrinsic cells is glutamatergic, acting through glutamate-gated chloride channels [32, 33], so the LPi15 label should be read as inhibitory of uncertain transmitter.

Table S7: Major shared inputs to the 100 LLPC1 cells.

Input	LLPC1 reached	Syn	NT	Proposed operation
LPi15	100	6,472	GABA	feed-forward opponent-direction inhibition
PVLP011	99	1,005	GABA	recurrent output gain control
Am1	99	648	GABA	wide-field modulatory inhibition
LPi13	85	499	GABA	lobula-plate inhibition
H1	78	1,111	Glut	heterolateral tangential context
VCH	77	588	GABA	direct wide-field inhibition plus T4/T5 gain control
LPT22	75	1,578	GABA	broad visual inhibition
H2	74	447	ACh	tangential context
LPi14	67	211	GABA	lateral/opponent lobula-plate inhibition
DCH	52	374	GABA	dorsal/parallel CH-like context

S8 Synapse-location (cable-distance) analysis

To test whether the three inhibitory operations act on the same or different parts of their target neurons, synapses were mapped onto the local FlyWire skeletons (node CSVs; the summary "status" flags are stale, but the skeletons carry full morphology, ∼0.9–2.3 mm cable). Each synapse was snapped to the nearest skeleton node and geodesic (cable) distances computed with `navis`. Usable skeletons exist for 286/454 T4a and 88/100 LLPC1; the conclusions are sign-identical under a Euclidean distance metric.

VCH terminal-gating on T4a. For each T4a, the cable distance from VCH→T4a input synapses to the nearest T4a→LLPC1 output synapse (median 2.3 μm) is far smaller than for the T4a's other inputs (median 51 μm); VCH is the closer input in 282 of 286 T4a (Wilcoxon $p = 7.5 \times 10^{-48}$). VCH therefore contacts the same T4a axon terminal that drives LLPC1 — the anatomy expected for presynaptic/terminal gain control rather than generic dendritic modulation.

LLPC1 input compartments. Cable distance from each inhibitory class to the nearest T4a-excitation synapse on the same LLPC1: LPi15 median 1.3 μm (IQR 1.0–1.9; $n = 88$), VCH 2.0 μm ($n = 47$), PVLP011 185 μm (IQR 164–212; $n = 86$). LPi15 and VCH are interdigitated with excitation on the dendritic input zone, whereas PVLP011 occupies a distinct distal compartment (Fig. S4c), consistent with recurrent output gating rather than dendritic input normalization. This is the dendrite-versus-axon distinction that the connectivity graph alone cannot resolve.

Table S8: Cable-distance synapse-location results (FAFB v783 skeletons).

Measurement	Median (μm)	n
VCH→T4a input → T4a→LLPC1 output terminal	2.3	286 T4a
Other T4a inputs → same output terminal	51.5	286 T4a
LPi15 → nearest T4a-excitation on LLPC1	1.3	88 LLPC1
VCH → nearest T4a-excitation on LLPC1	2.0	47 LLPC1
PVLP011 → nearest T4a-excitation on LLPC1	184.7	86 LLPC1

S9 VCH–T4/T5 normalization loop

The VCH–T4/T5 connectivity provides the first-stage normalization evidence. VCH receives 44.6% of its input synapses from T4/T5 and is dominated by T4a/T5a. VCH also sends GABAergic output back onto T4/T5. The 912 reciprocal T4/T5 partners are especially important: these neurons both drive VCH and receive VCH inhibition, providing the wiring motif for proportional feedback gain control.

Table S9: Key VCH–T4/T5 metrics.

Metric	Value
VCH neurotransmitter / class	GABA / visual centrifugal
Total VCH input / output synapses	27,576 / 32,363
T4/T5 inputs to VCH	1,022 neurons; 12,301 synapses
Dominant input subtypes	T4a 5,942 syn; T5a 6,225 syn
T4/T5 output targets of VCH	1,476 neurons; 7,273 synapses
Reciprocal T4/T5 partners	912 neurons
Fraction of VCH T4/T5 inputs also inhibited by VCH	89.1%
Mean input vs feedback synapses per T4/T5	12.0 vs 4.9
Dominant downstream sinks of reciprocal T4/T5	LPi14, VCH, CT1, HS family, DCH, LPi15

The reciprocal loop supports a gain-control interpretation rather than all-or-none gating. The excitatory drive to VCH is stronger than the inhibitory feedback to individual T4/T5 neurons, and VCH also targets subtypes beyond its strongest inputs. This suggests a broad, proportional adjustment of T4/T5 output under wide-field motion.

S10 Descending channels and motor mapping

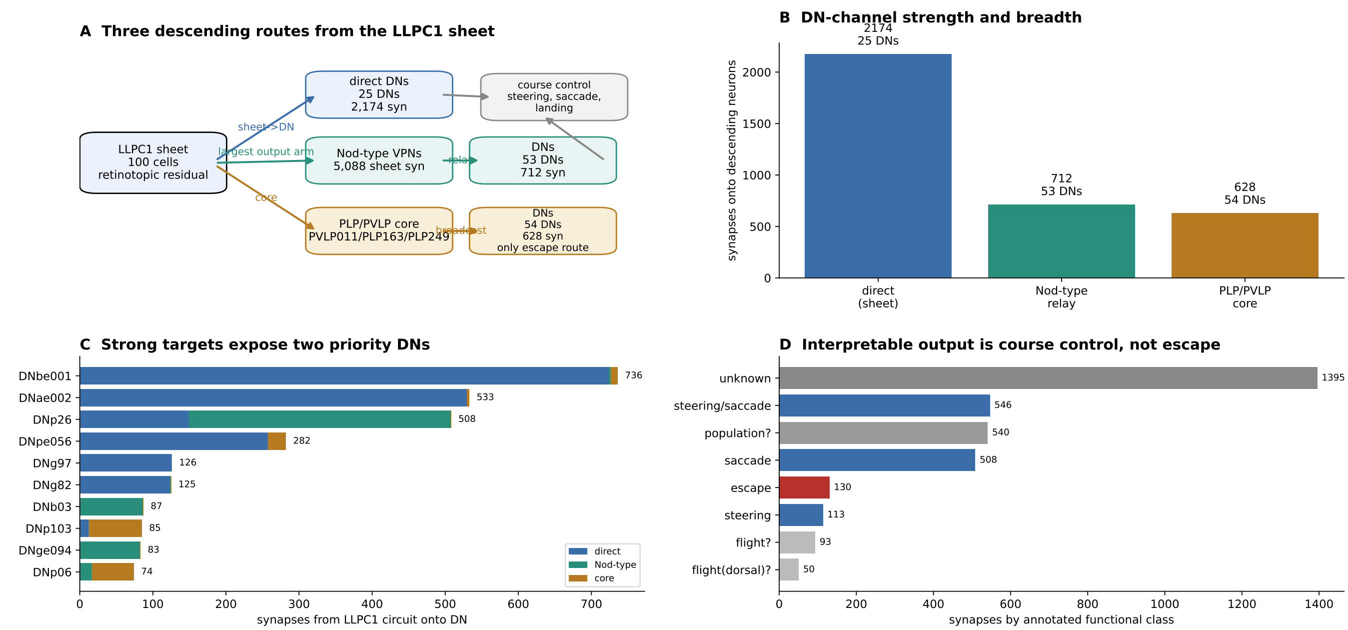

Figure S8: Descending routes before MCNS motor mapping. The DN enumeration divides LLPC1 output into direct sheet, Nod-type relay and PLP/PVLP broadcast-route channels. The direct channel carries the most synapses, the Nod arm is a strong second route led by DNp26, and the route is weak and broad but uniquely reaches looming/escape DNs.

S10.1 Nod-type split: only Nod1 relays to steering DNs

The Nod-type arm resolves into functionally distinct cells. We queried the descending output of each annotated Nod-type neuron driven by LLPC1 (direct, no-threshold synapse query of the annotated cells; the channel-level totals in the tables below come from the descending-neuron enumeration and may differ slightly owing to thresholding and the pooling of all Nod subtypes). Only the cholinergic Nod1 carries the figure signal to steering descending neurons.

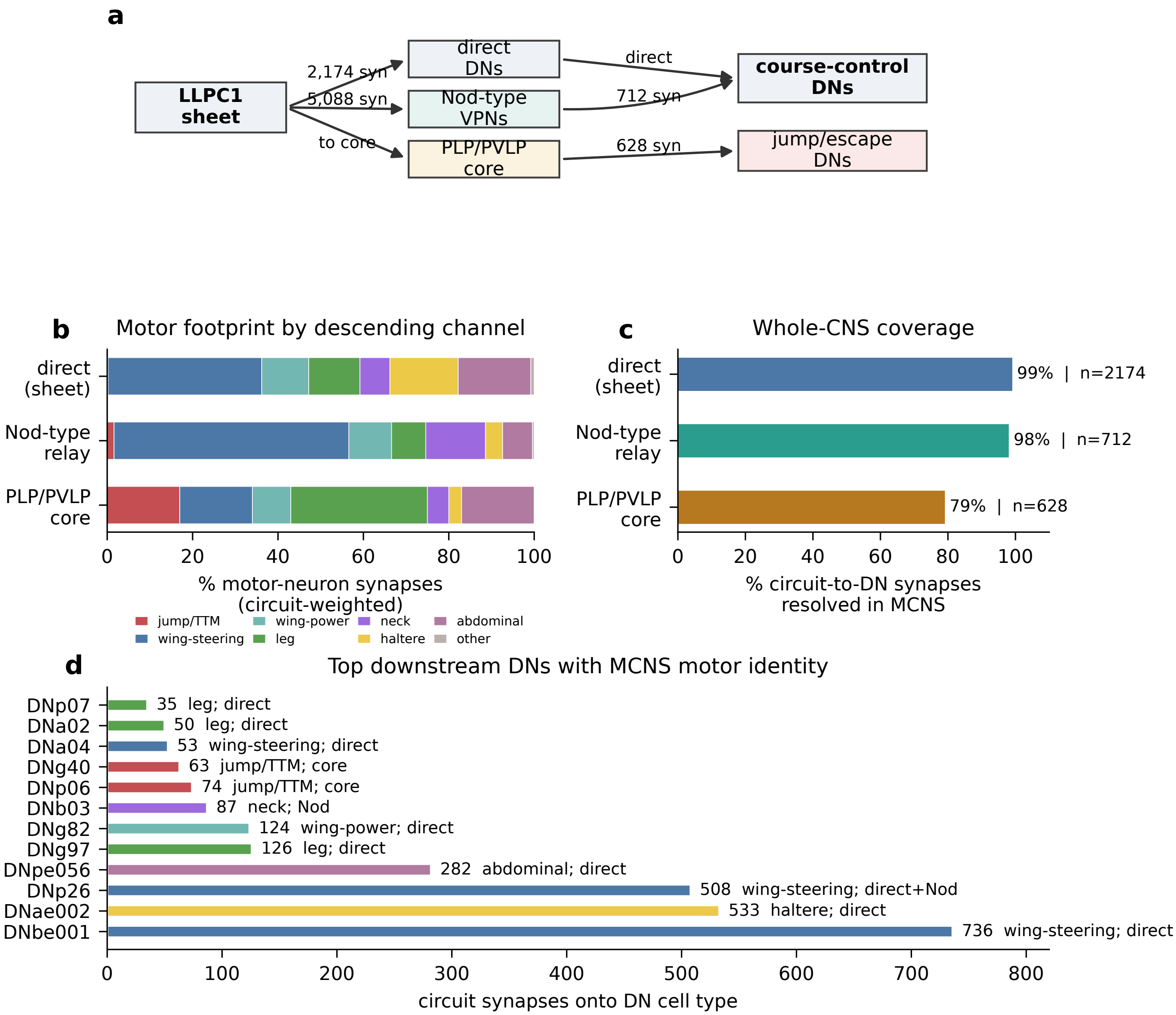

Figure S9: Whole-CNS motor mapping identifies the LLPC1 readout as wing-steering-led course control. a, Three-channel descending architecture. **b**, MCNS motor-neuron targets by channel. Direct and Nod-type routes are dominated by wing-steering and flight/gaze systems; the broadcast route is the only channel with substantial jump/TTM output. **c**, MCNS coverage is near complete for the two main figure-carrying routes. **d**, top DN targets with inferred motor identity. DNbe001 is the strongest direct target and resolves as wing-steering/flight-control; DNp26 is the strongest convergent direct+Nod target and also wing-steering-biased.

Table S10: Descending output of each Nod-type cell type driven by LLPC1. "LLPC1 syn" is the input the type receives from the sheet (number of the 100 LLPC1 contacting it in parentheses); "DN syn" is its total output onto descending neurons; "steering frac." is the fraction of that DN output reaching known steering DNs. Only the cholinergic Nod1 relays to the steering command neuron DNp26; the others are inhibitory (Nod2), route to neck/gaze (Nod5) or are weak.

Nod type	NT	LLPC1 syn	DN syn	→DNp26	steering frac.	top non-steering target
Nod1	ACh	4,228 (91)	662	**448**	644/662	—
LPT42_Nod4	ACh	255 (50)	216	106	123/216	DNg82 (wing-power)
Nod2	GABA	826 (70)	153	1	54/153	DNbe005
Nod3	ACh	21 (12)	90	0	14/90	DNge140
Nod5	ACh	13 (9)	333	0	0/333	DNb03 (neck/gaze, 247)

Table S11: Descending channel summary.

Channel	Circuit-to-DN synapses	DNs	MCNS coverage	Interpretation
Direct LLPC1→DN	2,174	25	99%	focused course-control route; strongest target DNbe001
Nod-type relay→DN	712	53	98%	major central-brain/LAL relay; led by convergent DNp26
PLP/PVLP broadcast route→DN	628	54	79%	weak broad broadcast; only substantial bridge to jump/escape

Table S12: MCNS motor-system distribution by descending channel. Percentages are circuit-weighted over resolvable targets with monosynaptic motor-neuron connections.

Channel	Jump	W-steer	W-power	Leg	Neck	Haltere	Abd.
Direct sheet	0	36	11	12	7	16	17
Nod-type arm	2	55	10	8	14	4	7
PLP/PVLP broadcast route	17	17	9	32	5	3	17

Table S13: Selected DN targets with MCNS motor identity.

DN	Circuit syn	Channel	Top motor system	Interpretation
DNbe001	736	direct	wing-steering	strongest target; wing-steering/flight-control footprint; no jump output
DNae002	533	direct	haltere	flight-balance output; direct course-control channel
DNp26	597	direct+Nod1	wing-steering	strongest convergent target; principal Nod1 steering output
DNpe056	282	direct	abdominal	flight-posture/body-axis output
DNg97	126	direct	leg	direct leg/walking-turn component
DNg82	124	direct	wing-power	flight-power-biased direct target
DNb03	87	Nod	neck	Nod arm neck/gaze component
DNp06	74	broadcast	jump/TTM	looming/escape-associated, route-only
DNg40	63	broadcast	jump/TTM	route-associated jump component
DNa04	53	direct	wing-steering	candidate flight-steering output
DNa02	50	direct	leg	walking steering command; central-complex output target
DNp07	35	direct	leg	visually evoked landing; graded leg extension

DN	Circuit syn	Channel	Top motor system	Interpretation
DNp01	16	broadcast	jump/TTM	Giant Fiber; route-only LLPC1 access

S10.2 Wing-specific steering control (MCNS motor-neuron side)

In the male whole-CNS connectome each motor neuron is annotated with the body side it innervates (`somaSide`). For the seven figure-driven wing-steering DN types we partitioned every DN cell's output onto wing-steering motor neurons (subclass `wm`, excluding the DLM/DVM power muscles) into the ipsilateral and contralateral wing. Six of the fourteen DN cells are strongly lateralized ($\geq$60% of steering output to one wing): the Nod1-driven DNp26 and DNg32 are contralateral, the directly driven DNa04 is ipsilateral, and DNbe001 is bilateral.

Table S14: Wing-steering laterality of the figure-driven descending neurons (MCNS). "Steering syn" is the DN type's total output onto wing-steering motor neurons (both cells); "ipsi. frac." is the fraction reaching the ipsilateral wing.

DN	figure input	steering syn	ipsi. frac.	wing target
DNa04	51 (direct)	908	0.99	ipsilateral
DNbe001	727 (direct)	1,181	0.52	bilateral
DNge107	22 (Nod1)	1,044	0.58	bilateral
DNbe005	30 (Nod1)	459	0.42	bilateral
DNp26	508 (Nod1)	386	0.23	**contralateral**
DNg32	54 (Nod1)	313	0.03	contralateral
DNge094	83 (Nod1)	37	0.00	contralateral

Table S15: DNp26 wing-steering motor-neuron targets in MCNS, by wing side relative to the DN. DNp26's strongest connections are onto the contralateral hg1 and i1 steering muscles; synapses are summed over both DNp26 cells.

steering muscle MN	wing (rel. to DN)	synapses
hg1 MN	contralateral	122
i1 MN	contralateral	105
hg2 MN	mixed (87 ipsi, 30 contra)	117
b3 MN	contralateral	28
hg3 MN	contralateral	5

S10.3 Per-channel MCNS target tables

The following tables reproduce the channel-specific motor identities that most directly support the main-text claim. "Circuit syn" denotes the connectome synapses onto that DN cell type across the relevant circuit routing, from our MCNS motor mapping. The direct and Nod-type routes are dominated by steering, balance and gaze-related motor systems. The PLP/PVLP route is the only channel with substantial jump/TTM output.

Table S16: Direct LLPC1→DN channel: leading MCNS-resolved targets.

DN	Circuit syn	Top motor system	Top motor-neuron targets / muscles
DNbe001	736	wing-steering	hg3 MN; DLMn c–f; hg1 MN
DNae002	533	haltere	MNhm42; MNnm03; b3 MN
DNpe056	282	abdominal	MNad01; ps2; MNad34
DNp26	219	wing-steering	hg1 MN; hg2 MN; i1 MN
DNg97	126	leg	tibia flexor; sternal anterior rotator
DNg82	124	wing-power	DLMn c–f; hg2 MN; DLMn a,b
DNa04	53	wing-steering	hg1 MN; MNnm08; b3 MN
DNa02	50	leg	sternal anterior rotator; femur reductor
DNp31	43	wing-power	DLMn c–f; MNwm36; ps1 MN

DN	Circuit syn	Top motor system	Top motor-neuron targets / muscles
DNp07	35	leg	tarsus levator; i2 MN; MNml81
DNge141	21	no monosynaptic MN	no monosynaptic MCNS motor-neuron target
DNb01	12	wing-steering	i1 MN; hg2 MN; MNad42

Table S17: Nod-type relay channel: leading MCNS-resolved targets.

DN	Circuit syn	Top motor system	Top motor-neuron targets / muscles
DNp26	289	wing-steering	hg1 MN; hg2 MN; i1 MN
DNb03	87	neck/gaze	ADNM1 MN; ADNM2 MN; b3 MN
DNge094	83	wing-steering	b3 MN; MNnm08; MNnm07/MNnm12
DNg32	54	wing-steering	MNhm43; tp1 MN; b2 MN
DNbe005	30	wing-steering	hg2 MN; hg1 MN; hg3 MN
DNge107	22	wing-steering	b1 MN; b2 MN; hg3 MN
DNge140	8	no monosynaptic MN	no monosynaptic MCNS motor-neuron target
DNb02	6	neck/gaze	MNhm03; MNnm11; ADNM1 MN
DNge141	5	no monosynaptic MN	no monosynaptic MCNS motor-neuron target
DNge135	4	wing-steering	tp2 MN; ps2 MN; MNad34
DNpe015	4	leg	MNhl88; MNhl87; hi2 MN
aSP22	3	leg	tergopleural/pleural promotor MN

Table S18: PLP/PVLP broadcast route: leading MCNS-resolved targets.

DN	Circuit syn	Top motor system	Top motor-neuron targets / muscles
DNp103	85	leg	sternal anterior rotator MN; TTMn
DNp06	74	jump/TTM	TTMn; MNad42; sternal anterior rotator
DNg40	63	jump/TTM	TTMn; tergopleural/pleural promotor MN
DNp35	50	no monosynaptic MN	no monosynaptic MCNS motor-neuron target
DNae007	36	no monosynaptic MN	no monosynaptic MCNS motor-neuron target
DNa07	30	wing-steering	hg1 MN; b2 MN; DLMn c–f
DNp27	28	abdominal	ps1 MN; hg2 MN; MNad02
DNpe037	24	wing-power	DVMn 1a–c; DVMn 3a,b; DVMn 2a,b
DNp11	22	leg	MNad34; sternal posterior rotator MN
DNp03	19	wing-power	DLMn c–f; ps1 MN; MNwm36
DNp04	18	no monosynaptic MN	no monosynaptic MCNS motor-neuron target
DNa16	17	haltere	MNhm03; ADNM1 MN; hg4 MN
DNp01	16	jump/TTM	TTMn; DVMn 1a–c

S10.4 DNbe001 and DNp26 as primary functional endpoints

DNbe001 and DNp26 are the two most informative first targets for physiology. DNbe001 is the strongest direct downstream target: it receives input from 70 of the 100 LLPC1 cells and has 736 total circuit synapses in the MCNS table. It makes 2,302 monosynaptic motor-neuron synapses in MCNS, dominated by wing-steering output (1,181 synapses), with additional abdominal, wing-power, haltere and neck targets and no leg or jump output. Its "wing-steering" assignment reflects the synapse-weighted majority; the prominent dorsal-longitudinal (DLM) targets add a flight-power component. DNbe001 is therefore anatomically resolved as a wing-steering/flight-control descending neuron, although its behaviour has not been characterized directly.

DNp26 is the strongest convergent readout. It is reached through both the direct sheet route and the Nod1 relay; a direct, no-threshold query gives 448 synapses from Nod1 and a further 149 directly from the sheet, so Nod1 supplies the majority of its figure drive. MCNS resolves DNp26 as wing-steering-biased, with prominent hg1, hg2 and i1 wing-steering motor targets. This makes DNp26 the best target for testing whether the Nod-type arm carries an amplified, temporally filtered or spatially pooled copy of the LLPC1 motion residual.

S10.5 High-mass MCNS motor identities

The table below reproduces the high-mass portion of the MCNS motor readout. It is not used as a statistical test in the main text; its purpose is to make the motor interpretation auditable. DNbe001, DNae002 and DNpe056 are especially important because they were unresolved or poorly resolved in VNC-only MANC but are resolved in whole-CNS MCNS.

Table S19: MCNS-resolvable DN targets sorted by monosynaptic motor-neuron synapses (high-mass subset).

DN	MN syn	Top motor system	Top motor targets / muscles
DNp31	4,071	wing-power	DLMn c–f; MNwm36; ps1
DNbe001	2,302	wing-steering	hg3; DLMn c–f; hg1
DNg82	1,888	wing-power	DLMn c–f; hg2; DLMn a,b
DNa02	1,501	leg	sternal anterior rotator; femur reductor
DNp03	1,284	wing-power	DLMn c–f; ps1; MNwm36
DNa04	1,266	wing-steering	hg1; MNnm08; b3
DNa10	1,253	wing-steering	i2; b3; i1
DNp15	1,213	neck	MNnm11; ADNM1; MNnm14
DNge107	1,182	wing-steering	b1; b2; hg3
DNa06	1,154	neck	ADNM1; MNnm11; MNnm14
DNae002	945	haltere	MNhm42; MNnm03; b3
DNg02_f	928	wing-power	DLMn c–f; MNwm36; ps1
aSP22	921	leg	tergopleural/pleural promotor
DNa09	892	haltere	MNhm42; MNhm43; MNnm03
DNb05	884	leg	sternotrochanter; tibia extensor
DNae003	883	haltere	MNhm42; MNhm43; MNnm03
DNg32	849	wing-steering	MNhm43; tp1; b2
DNa16	825	haltere	MNhm03; ADNM1; hg4
DNpe005	811	wing-power	hg2; DVMn 1a–c; DVMn 2a,b
DNp63	801	wing-steering	hg2; hg1; hDVM
DNb01	661	wing-steering	i1; hg2; MNad42
DNge030	655	wing-power	DLMn c–f; MNad42; DLMn a,b
DNp26	619	wing-steering	hg1; hg2; i1
DNbe005	614	wing-steering	hg2; hg1; hg3
DNp13	589	abdominal	ps2; MNhl59; MNad53
DNp11	586	leg	MNad34; sternal posterior rotator
DNp102	584	haltere	MNhm42; MNhm43; sternal anterior
DNp07	516	leg	tarsus levator; i2; MNml81
DNg97	423	leg	tibia flexor; sternal anterior rotator
DNa07	421	wing-steering	hg1; b2; DLMn c–f
DNae004	388	wing-steering	i1; MNhm42; b3
DNp47	386	leg	tergopleural/pleural promotor
DNb02	352	neck	MNhm03; MNnm11; ADNM1
DNpe045	331	abdominal	MNad34; MNad63; hDVM
DNpe037	303	wing-power	DVMn 1a–c; DVMn 3a,b; DVMn 2a,b
DNa13	240	leg	sternal posterior rotator
DNpe021	239	abdominal	iii1; MNad16; MNad19
DNge092	227	wing-steering	MNnm03; hg4; MNwm35
DNb03	186	neck	ADNM1; ADNM2; b3
DNpe013	179	leg	pleural remotor/abductor; tarsus levator
DNg34	139	jump/TTM	tergotrochanter; sternotrochanter
DNp64	128	abdominal	MNad06; DVMn 1a–c; ps2
DNp06	126	jump/TTM	TTMn; MNad42; sternal anterior rotator
DNp54	120	wing-steering	tp2; DVMn 1a–c; ps2
DNpe015	105	leg	MNhl88; MNhl87; hi2
DNg46	102	neck	MNnm13; MNnm10; MNnm03

DN	MN syn	Top motor system	Top motor targets / muscles
DNp01	94	jump/TTM	TTMn; DVMn 1a–c
DNp09	93	abdominal	tpn; MNad34; MNml81

S11 Alternative circuits: full evaluation

Table S20: Candidate alternatives and why they are not sufficient as complete models.

Candidate	Strength	Problem	Final assignment
HS family	Direct T4/T5 input, wide-field motion response and optomotor output [26, 28, 29]; historical large-field candidate	Collapses retinotopy and is tuned for whole-field stabilization; cannot be first position-coded residual sheet	Wide-field stabilization and context output
VCH	GABAergic; massive T4/T5 input; reciprocal feedback to 912 T4/T5 partners; historical CH/FD precedent	Broad normalizer rather than excitatory figure representation; does not by itself route a spatial map to DNs	First-stage normalization/context
DCH	CH-like centrifugal GABAergic cell; dorsal/parallel context	Weaker evidence than VCH for figure-selective causal role; not a sheet output	Parallel wide-field context
LPi15	Universal inhibitory input to LLPC1; strong opponent-direction drive [32, 33]	Mainly feed-forward; not a recurrent output map and not a motor route	Opponent-direction sheet inhibition
LPi14	Dominant T4/T5 downstream inhibitory sink; likely lateral/opponent suppression	Sparse as common LLPC1 input compared with LPi15; no full readout to DNs	Lobula-plate inhibition/opponency
PVLP011	Reads all LLPC1 and inhibits 99/100 back; recurrent motif	GABAergic normalizer; not the main descending route	Recurrent gain control
PLP163	Cholinergic, contacted by all 100 LLPC1	Weaker than direct/Nod routes to DNs; not sufficient as bottleneck	Excitatory readout
PLP249	Strong GABAergic broadcast-route target	Selector/gate; reaches escape through the broadcast route but not primary course-control route	Broadcast-route selector and escape-interface gate
Nod-type cells	Largest intermediate LLPC1 output arm; relays to 53 DNs led by DNp26	Downstream of LLPC1; nodulus identity uncertain; not first figure cell	Central-brain/LAL course-control relay
LC/T3 object pathways	Important for object, bar and target responses; reaches optic glomeruli [46–48, 65]	Parallel feature/action systems; do not explain T4/T5-normalized LLPC1 residual	Parallel object channels
LPLC2/LC4 looming	Strong looming/escape pathways and jump DNs [44, 45]	LLPC1 direct/Nod routes avoid jump; escape appears only through the weak broadcast route	Parallel collision/escape system
DNs	Direct motor access; DNbe001 and DNp26 are key endpoints	Too late to explain retinotopic extraction and normalization	Behavioral readout

Table S21: Criteria-based candidate screen. Each candidate is scored on the six requirements for a *first* retinotopic motion-residual sheet. Only LLPC1 satisfies all six. Y = yes, P = partial, N = no; "–" = not applicable (the cell is an inhibitory or relay element, not a candidate output sheet).

Candidate	local T4/T5	retinotopy	shared inhib.	recurrence	direct DN	steer $\neq$ jump
LLPC1	Y (15/cell)	Y (z=−68)	Y	Y	Y	Y
HS family	Y (∼839/cell)	N (wide-field)	P	P	Y	Y (optomotor)
VCH	Y (whole-field)	N	Y (norm.)	Y	N	N
LPi15	Y (layer-b)	N	Y (supplies)	N (feed-fwd)	N	N
PVLP011	N (0 T4/T5)	N	Y	Y	P	P
Nod-type relay	N (relay)	N	–	–	Y	Y
PLP/PVLP route	N	N	–	–	Y (broad)	N (jump/mixed)
LPLC2/LC4	P	N (glomeruli)	–	–	Y	N (escape)
LC/T3	P (feature)	P	–	–	Y	P (object)

S12 A minimal sheet-readout model

A minimal sheet model can be written as

$$L_i(t) = \phi\left(E_i(t) - \alpha I_i^{\mathrm{opp}}(t)\right), \tag{1}$$

where E_i is local T4a/T5a drive to LLPC1 cell i, I_i^{opp} is LPi15/opponent inhibition, and ϕ is a rectifying or sigmoidal synaptic nonlinearity. VCH/DCH–HS context and PVLP011 recurrence then set the gain:

$$F_i(t) = \frac{L_i(t)}{\beta + \gamma P_{\mathrm{VCH}}(t) + \delta P_{\mathrm{PVLP011}}(t)}. \tag{2}$$

Under coherent panorama motion, local drive and the denominator rise together, so F_i is compressed across the sheet. Under local figure motion, a subset of E_i differs from the global context term and survives as a residual. The direct and Nod-type readouts can then implement weighted sums over the retinotopic residual:

$$D_k(t) = \sum_i w_{ki} F_i(t) + \eta_k C(t), \tag{3}$$

where D_k is a downstream DN or central-brain relay and $C(t)$ denotes state-dependent context. The model predicts that DNbe001 and DNp26 should track residual figure motion more strongly than raw optic-flow energy.

S13 Experimental predictions

LLPC1 imaging. LLPC1 should respond more strongly to local motion incoherence than to coherent whole-field motion of matched local contrast. Responses should be spatially tiled across the sheet rather than HS-like whole-field signals.

VCH perturbation. Silencing or hyperpolarizing VCH should increase T4/T5 and LLPC1 gain under wide-field motion and reduce the contrast between local residual and global background. Artificial VCH activation should divisively suppress LLPC1 responses.

LPi15 perturbation. Silencing LPi15 should reduce opponent-direction suppression and make LLPC1 more responsive to raw motion energy, especially opponent horizontal motion that normally suppresses the sheet.

PVLP011 perturbation. PVLP011 silencing should reduce recurrent normalization or persistence and alter the temporal stability of sheet responses after transient figure motion.

Descending readout. DNbe001 and DNp26 should encode figure-motion residuals and drive wing-steering or flight-posture changes. Their activation should bias steering or course correction more readily than jump. Broadcast-route-biased activation should be more likely to modulate escape thresholds or reach jump/TTM output.

Behavioral dissociation. Figure-with-moving-ground assays should be more sensitive to LLPC1/direct/Nod perturbation than looming assays. LPLC2/LC4 perturbation should affect looming escape without abolishing the LLPC1 course-control residual.

S14 Limitations

The main limitations are anatomical and functional. Connection counts are not synaptic weights; the connectome omits electrical synapses; neurotransmitter labels are predictions except where experimentally established; motor mapping uses a male CNS while the source brain is female FAFB; and MCNS counts only monosynaptic DN-to-motor connections, so DNs acting through VNC interneurons are undercounted. The circuit is also one direction and one hemisphere exemplar. Most importantly, the connectome establishes the anatomical *requirements* for a normalized local motion residual — local T4/T5 excitation, shared opponent and recurrent inhibition, and direct course-control output — not the response property itself; demonstrating residual selectivity requires physiology in LLPC1 and its descending targets. These caveats do not remove the central result, because the main conclusion depends on several independent signals: local LLPC1 retinotopy, shared inhibitory architecture, three-channel DN routing, and MCNS motor identity of the downstream routes.

S15 Data and reproducibility statement

All analyses use public connectomic resources. FlyWire-derived analyses used materialization v783 of `flywire_fafb_public`, with chemical synapses queried from the CAVE API (`synapses_nt_v1`) using all synapses and no cleft-score threshold. The descending-channel analysis records all direct sheet-to-DN, Nod-type-relay-to-DN and PLP/PVLP-broadcast-to-DN synapses with per-cell annotations. The motor read-out cross-matched the downstream DN cell types into the male CNS connectome (`male-cns:v1.0`) and classified their monosynaptic motor-neuron targets by curated motor-neuron subclass. Analysis code and intermediate tables are available from the authors.

S16 Evidence-to-claim ledger

The central claim is supported by four independent constraints rather than by any single edge count. The sheet is local and retinotopic; its inhibition is broad and layered; its output is routed to motor systems through identified DNs; and the alternatives each fail at least one of these constraints. This ledger is useful because it separates anatomy that is directly counted from inference that requires physiology.

Table S22: How the main claims map onto specific evidence.

Claim	Primary support	Remaining test
LLPC1 is the first retinotopic figure-motion sheet	454 VCH-targeted T4a converge onto 100 right LLPC1 at local scale; T4a/T5a dominate excitation	Image LLPC1 under matched figure-ground phase and size conditions
The sheet computes a residual, not raw motion energy	Local excitation is paired with LPi15, VCH/DCH–HS and PVLP011 inhibition	Perturb LPi15, VCH and PVLP011 separately and test residual selectivity
VCH presynaptically gates the T4 sites driving LLPC1	VCH→T4a synapses sit a median 2.3 μm (cable) from the T4a→LLPC1 output terminal vs 51 μm for other inputs (99% of 286 T4a)	Test whether VCH silencing changes T4a terminal gain onto LLPC1
Sheet inhibition is compartmentalized (dendrite vs axon)	Cable distance to nearest T4a excitation on LLPC1: LPi15 1.3, VCH 2.0, PVLP011 185 μm	Compartment-specific perturbation — PVLP011 should change output gain, LPi15 should change tuning
Direct and Nod-type routes carry course-control output	DN enumeration plus MCNS motor map: direct/Nod are wing-steering-led, gaze/haltere/posture-biased	Image and silence DNbe001 and DNp26 during figure tracking in flight
The PLP/PVLP route is not the main steering bottleneck	It has fewer circuit-to-DN synapses, mixed motor output and the only substantial jump/TTM footprint	Test whether broadcast-route-biased activation modulates escape thresholds rather than steering alone
LPLC2/LC4 and HS are parallel systems	LLPC1 direct/Nod output avoids jump, whereas HS collapses retinotopy	Double perturbation with looming and figure-ground stimuli

S17 A two-stage normalization model

The circuit is naturally written in two stages. In *stage 1* the separation is performed by VCH at the T4/T5 terminal. Let $x_i^a(t)$ be the raw local T4a/T5a motion drive at the terminal feeding LLPC1 cell i, and $S^a_{\mathrm{VCH}}(t)$ the wide-field same-direction T4/T5 pool that drives VCH. VCH feedback gates the terminal output divisively,

$$\tilde{x}_i^a(t) = \frac{x_i^a(t)}{\beta + \gamma\, S^a_{\mathrm{VCH}}(t)}. \tag{4}$$

Coherent panoramic motion drives S^a_{VCH} and is broadly suppressed, whereas local motion that does not strongly recruit the wide-field pool survives; the residual-like signal $\tilde{x}_i^a$ is therefore formed *before* LLPC1. In *stage 2*, LLPC1 cell k reads out the gated signal over its receptive field and is regulated by the sheet inhibitors,

$$L_k(t) = \phi\Big(\textstyle\sum_{i\in \mathrm{RF}_k} w_{ki}\, \tilde{x}_i^a(t) \;-\; \lambda\, I_k^{\mathrm{LPi15}}(t) \;-\; \delta\, I_k^{\mathrm{PVLP011}}(t)\Big), \tag{5}$$

where I^{LPi15} is sheet-wide (operating-point) inhibition and I^{PVLP011} is recurrent output regulation. Stage 2 tunes how the sheet is stabilized, suppressed or routed, but the minimal background-removal operation has already occurred in stage 1. The central physiological prediction is that LLPC1 and its direct/Nod outputs encode the VCH-gated residual $\tilde{x}_i^a$ more strongly than raw motion energy, and that removing VCH — not LPi15 or PVLP011 — is what abolishes the separation.

S18 Priority experiments

The strongest immediate tests are not generic LLPC1 silencing experiments, but matched perturbations across the three operations and two leading descending readouts. These experiments would turn the connectomic model into a causal circuit model.

Table S23: Priority experiments for a first-pass causal test.

Target	**Assay**	**Prediction**	**Critical control**
LLPC1	two-photon imaging during figure-with-moving-ground stimuli	spatially localized residual; weak coherent-background response	compare with HS under identical stimuli
VCH	acute silencing/activation	changes wide-field gain of T4/T5 and LLPC1; activation divisively suppresses residual	local figure on static background
LPi15	silencing	reduces opponent-direction suppression and increases raw motion-energy responses in LLPC1	direction-specific T4/T5 stimuli
PVLP011	silencing	reduces recurrent gain control or persistence across the sheet	LPi15-silencing comparison
DNbe001	imaging and activation in tethered flight	encodes residual and biases wing steering without jump	looming/TTM assay
DNp26	image direct vs Nod-route perturbations	convergent residual readout; possible saccade/wing-steering bias	block Nod-type relay while preserving direct LLPC1 input

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